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Comparative neocortical neuromorphology in felids: African lion, African leopard, and cheetah

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Abstract

The present study examines cortical neuronal morphology in the African lion (*Panthera leo leo*), African leopard (*Panthera pardus pardus*), and cheetah (*Acinonyx jubatus jubatus*). Tissue samples were removed from prefrontal, primary motor, and primary visual cortices and investigated with a Golgi stain and computer-assisted morphometry to provide somatodendritic measures of 652 neurons. Although neurons in the African lion were insufficiently impregnated for accurate quantitative dendritic measurements, descriptions of neuronal morphologies were still possible. Qualitatively, the range of spiny and aspiny neurons across the three species was similar to those observed in other felids, with typical pyramidal neurons being the most prominent neuronal type. Quantitatively, somatodendritic measures of typical pyramidal neurons in the cheetah were generally larger than in the African leopard, despite similar brain sizes. A MARsplines analysis of dendritic measures correctly differentiated 87.4% of complete typical pyramidal neurons between the African leopard and cheetah. In addition, unbiased stereology was used to compare the soma size of typical pyramidal neurons (n = 2,238) across all three cortical regions and gigantopyramidal neurons (n = 1,189) in primary motor and primary visual cortices. Both morphological and stereological analyses indicated that primary motor gigantopyramidal neurons were exceptionally large across all three felids compared to other carnivores, possibly due to specializations related to the felid musculoskeletal systems. The large size of these neurons in the cheetah which, unlike lions and leopards, does not belong to the *Panthera* genus, suggests that exceptionally enlarged primary motor gigantopyramidal neurons evolved independently in these felid species.

KEYWORDS

cerebral cortex, dendrites, gigantopyramidal neuron, Golgi stain, pyramidal neuron, RRID: SCR_001775, RRID:SCR_001905, RRID:SCR_002526

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1 | INTRODUCTION

In the last decade, the neocortical neuronal morphology of relatively rare large-brained mammals has been explored within a comparative framework (African elephant: Jacobs et al., 2011; cetaceans: Butti et al., 2015; giraffe: Jacobs, Harland, et al., 2015). The two most recent studies in this series examined nondomestic feliforms. Johnson et al. (2016) explored a variety of cortical neuron types in the Siberian tiger and clouded leopard, noting that, consistent with Brodmann's (1909) initial observations, gigantopyramidal neurons in carnivore primary motor cortex (M1) were extremely large. Subsequently, Jacobs et al. (2018) examined both typical pyramidal and gigantopyramidal neurons in M1 using two methodologies: a Golgi technique to provide qualitative and quantitative somatodendritic measures across 19 mammalian species from 7 orders, including several feliforms (e.g., African lion, caracal, clouded leopard, and Siberian tiger), and unbiased stereology to compare soma size across 11 carnivores (including African leopard, African lion, caracal, and Siberian tiger) and 9 primate species. Both analyses revealed that M1 gigantopyramidal neurons were larger in feliforms (especially in the *Panthera* species) than in other (sub)orders, presumably due to specializations of the felid neuromusculoskeletal systems. The present study is thus a direct extension of both Johnson et al. (2016) and Jacobs et al. (2018). Employing measurements of both dendritic morphology and soma size, the present study examines neocortical neuron types in the prefrontal cortex, M1, and primary visual cortex (V1) of the African lion (*Panthera leo leo*), the African leopard (*Panthera pardus pardus*), and the cheetah (*Acinonyx jubatus jubatus*), with stereological results also compared to other species of Carnivora.

These three members of the family Felidae emerged from the order Carnivora in the late Oligocene epoch ~35 million years ago (Ma; Kleiman & Eisenberg, 1973). Both the African lion and African leopard are members of the *Panthera* genus, the first of eight clades within the felids to diverge ~10.8 Ma (Janczewski, Modi, Stephens, & O'Brien, 1995; O'Brien & Johnson, 2005). Moreover, phylogenetic analyses (chemical signaling: Bininda-Emonds, Decker-Flum, & Gittleman, 2001; mitochondrial DNA control region: Jae-Heup, Eizirik, O'Brien, & Johnson, 2001; genomic assessment: Yu & Zhang, 2005) suggest that they are sister taxa, with the African lion being one of the largest extant felids (males: ~187 kg; females: ~124 kg; Smuts, Robinson, & Whyte, 2009). The ~50 kg African leopard is a descendant of an early branch in the clade and the most prominent subspecies of the six geographically isolated groups that compose the leopard lineage (*Panthera pardus*; Miththapala, Seidensticker, & O'Brien, 1996; Uphyrkina et al., 2001). In contrast, the cheetah originated from the North American puma lineage that diverged much later (~6.7 Ma) and remained isolated during a >100,000 year-old Pleistocene migration across the Beringian land bridge from North America to central Asia and Africa (Barnett et al., 2005; Durant et al., 2017; O'Brien & Johnson, 2005). The cheetah, characterized by a very low genetic polymorphism across the genome (Dobrynin et al., 2015), exhibits unique osteological characteristics more similar to caniforms than to felines, greatly differentiating it from other extant felids (Kleiman & Eisenberg, 1973; Russell & Bryant, 2001). This atypical felid is thus an important comparison point for neuromorphological analysis.

Qualitatively, previous neuromorphological findings have documented considerable similarity across both domestic (Hübener, Schwarz, & Bolz, 1990; Kaiserman-Abramof & Peters, 1972; Matsubara, Chase, & Thejomayen, 1996; Winer, 1984) and nondomestic (Jacobs et al., 2018; Johnson et al., 2016) felids. Therefore, we expect neocortices in the present sample to exhibit predominantly vertically oriented typical pyramidal neurons, along with several other neuronal types (e.g., extraverted pyramidal, gigantopyramidal, aspiny neurons). Although quantitative differences in regional dendritic extent have been documented for pyramidal neurons in primate (Bianchi et al., 2013; Elston & Rockland, 2002; Jacobs et al., 2001) and African elephant (Jacobs et al., 2011) neocortex, similar regional differences have not been observed in the Siberian tiger or the clouded leopard (Johnson et al., 2016). As such, there is no a priori reason to expect such differences in the current sample. In addition, cross-species comparisons suggest that increased brain size is positively correlated with greater somatodendritic measures within mammalian lineages (Butti et al., 2015; Elston & Manger, 2014; Jacobs, Harland, et al., 2015; Johnson et al., 2016). The average brain mass of the African lion (~220 g) is twice that of both the cheetah (~110 g) and the African leopard (~120 g; Gittleman, 1986). Therefore, we expected that the African lion neurons would be more dendritically complex than those from the other two felids in the present study. Unfortunately, because of the suboptimal quality of the Golgi stain in the lion tissue, only a qualitative description of these dendritic systems was possible. However, these descriptive results still broaden the neuromorphological findings of Jacobs et al. (2018), which focused only on pyramidal and gigantopyramidal neurons in M1 cortex. Finally, soma size comparisons across all three species were possible with data from unbiased stereological analyses.

Both Golgi and stereological techniques also provided size comparisons of gigantopyramidal neuron somata across the three species. Given previous findings for *Panthera*, M1 gigantopyramidal neurons in both the African lion and African leopard were expected to be exceptionally large relative to other carnivore species, potentially due to their specialized hunting behaviors (Jacobs et al., 2018; Johnson et al., 2016). However, the somatodendritic characteristics of M1 gigantopyramidal neurons in the cheetah have not been documented previously. The morphology of these neurons is of particular interest because the cheetah is a distant non-*Panthera* felid that exemplifies extraordinary running speed, reaching sprints of >100 km/h when catching prey (Sharp, 1997). To achieve this, cheetahs exhibit a modified morphology compared to other felids, having elongated hind legs and spine, an aerodynamic skull, semiretractable claws, and enhanced metabolic and adrenal systems (Hudson et al., 2010, 2011; Russell & Bryant, 2001). Further, the cheetah's specialized vestibular system is larger than that of other felids, allowing it to maintain postural stability and balance during prey pursuits (Grohé, Lee, & Flynn, 2018). The cheetah has evolved to prioritize acceleration and agility over the pouncing behavior characteristic of *Panthera* (Kleiman & Eisenberg, 1973). Moreover, it has a muscle fiber composition designed for extremely fast sprints of limited duration (Goto et al., 2013). As such,

specialized gigantopyramidal neuron morphology may be needed to support the cheetah's fine motor control. Thus, another purpose of the present study was to examine qualitative and quantitative characteristics of M1 gigantopyramidal neurons in the cheetah.

2 | MATERIALS AND METHODS

2.1 | Specimens

Cortical tissue of three feline species was collected from four different specimens: two African lions from the Copenhagen Zoo in Denmark (adult male, brain mass = 166.4 g; adult female, 184.5 g), one African leopard from the Maryland Zoo in the U.S. (22-year-old male, brain mass = 143.2 g), and one cheetah from the Borås Zoo in Sweden (adult female, brain mass = 139.5 g). All brains were in situ perfusion-fixed (autolysis time, AT <1 hrs) via the carotid artery with cold saline followed by 4% paraformaldehyde in 0.1 M phosphate buffer. Following removal, brains were stored in 4% paraformaldehyde in 0.1 M phosphate buffer for 24 hr. Selected tissue blocks were stored in 0.1% sodium azide in 0.1 M phosphate-buffered saline at 4°C for different time periods (lions: 27 months; leopard: 6 months; cheetah: 18 months) prior to Golgi staining. The present study was approved by the Colorado College Institutional Review Board (#011311-1) and the University of the Witwatersrand Animal Ethics Committee (#2012/53/1), which comply with NIH policies for the care and use of animals in scientific experimentation.

2.2 | Tissue selection and preparation

Tissue blocks (3–5 mm thick) were removed from three regions of the left hemisphere in all specimens: (a) the dorsomedial aspect of the frontopolar cortex; (b) M1, located immediately posterior to the frontal pole and 1–2 cm lateral to the longitudinal fissure; and (c) V1, located along the dorsal posterior aspect of the lateral gyrus and 1–2 cm lateral to the longitudinal fissure (Figure 1). Tissue was coded to prevent experimenter bias, stained with a modified rapid Golgi technique (Scheibel & Scheibel, 1978), and serially sectioned at 120 μm with a Vibratome (Leica VT1000S, Leica Microsystems, Wetzlar, Germany).

2.3 | Neuron selection and quantification

Neurons were chosen for quantification according to established criteria (Jacobs et al., 2011; Jacobs et al., 2018; Jacobs, Harland, et al., 2015; Jacobs, Lee, et al., 2015; Johnson et al., 2016), with an isolated soma near the center of the 120-μm-thick section and relatively well impregnated, unobscured, and complete (i.e., nontruncated) dendritic projections. To provide a comprehensive morphological analysis, neurons were selected to encompass representative typologies across all three species. Neurons were quantified under a planachromatic 60X oil objective along *x*-, *y*-, and *z*-coordinates using a Neurolucida system (v2018.2.2, MBF Bioscience, Williston, VT; RRID: SCR_001775), interfaced with an Olympus BH-2 microscope equipped with a Ludl XY motorized stage (Ludl Electronics, Hawthorne, NY) and a Heidenhain *z*-axis encoder (Schaumburg, IL). A MicroFire Digital CCD 2-Megapixel camera (Optronics, Goleta, CA) mounted on a trinocular head (model 1-L0229, Olympus, Center Valley, PA) displayed images on a 1,920 × 1,200 resolution Dell E248WFP 24-in. LCD monitor. Somata were first traced at their widest point in a two-dimensional plane to estimate cross-sectional areas. Dendrites were then traced somatofugally, accounting for dendritic diameter and quantity of spines. Dendritic arbors were not followed into neighboring sections; broken ends and indefinite terminations were labeled as incomplete endings. Neurons with sectioned segments were not differentially analyzed because elimination of such tracings would have biased the sample toward smaller neurons (Uylings, Ruiz-Marcos, & Pelt, 1986).

Intrarater reliability was evaluated by having the four investigators (V.T.N., R.U., A.W., L.J.S.) trace the same soma, dendritic segment, and spines 10 times. The average coefficient of variation for soma size

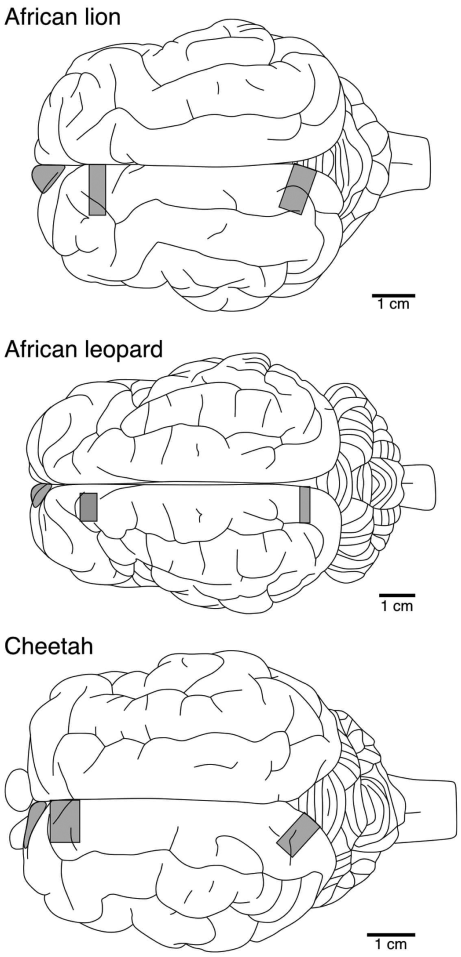

FIGURE 1 Dorsal schematic views of the African lion, African leopard, and cheetah brains. Highlighted are the relative positions of the cortical sections sampled from anterior to posterior: prefrontal cortex, primary motor cortex, and primary visual cortex. Note that brains are scaled differently from each other. Scale bar = 1 cm

(3.9%), total dendritic length (TDL, 1.9%), and dendritic spine number (DSN, 4.4%) revealed minimal variation in tracings. A split plot design (α = .05) indicated no significant difference between the first and last five tracings of the data set. Interrater reliability was evaluated by comparing 10 tracings of different dendritic systems between each investigator and the primary investigator (B.J.). Interclass correlation averages for soma size (1.0), TDL (0.98), and DSN (0.98) were not significantly different between investigators (Analysis of Variance, ANOVA, α = .05). Additionally, the primary investigator reexamined all completed tracings under the microscope to ensure accuracy.

2.4 | Quantitative dendritic and spine measures

Neurons were classified according to somatodendritic criteria, including soma size and shape, presence of spines, laminar location, and general morphology (Betz, 1874; Ferrer, Fabregues, & Condom, 1986a, 1986b; Jacobs et al., 2011; Jacobs et al., 2018; Johnson et al., 2016). Dendritic systems were classified centrifugally (Bok, 1959; Uylings et al., 1986). Soma size (i.e., surface area, μm^2) and soma depth from the pial surface (μm) were quantified, as well as six other well-established measures (Jacobs et al., 2011; Jacobs et al., 2018; Johnson et al., 2016): dendritic volume (Vol, μm^3: the total volume of all dendrites); total dendritic length (TDL, μm: the summed length of all dendritic segments); mean segment length (MSL, μm: the average length of each dendritic segment); dendritic segment count (DSC: the number of dendritic segments); dendritic spine number (DSN: the total number of spines on dendritic segments); and dendritic spine density (DSD: the average number of spines per μm of dendritic length). Dendritic branching patterns were also analyzed using a Sholl analysis (Sholl, 1953), which quantified dendritic intersections at 20-μm intervals radiating somatofugally. All descriptive measures are presented as mean ± standard error of the mean (*SEM*) unless noted otherwise.

2.5 | Statistical analysis of dendritic differences across species

A total of 652 neurons was traced across the three felid species: African lion (162; 81 from each animal), African leopard (257), and cheetah (233). Neurons were traced in prefrontal cortex (106; lion = 20, leopard = 35, cheetah = 51), M1 (327; lion = 66, leopard = 151, cheetah = 110), and V1 (219; lion = 76, leopard = 71; cheetah = 72). Tracings from both African lions were excluded from statistical analyses because of the relatively incomplete Golgi stain. To decrease artifactual variability, only the numerically dominant typical pyramidal neurons were examined. Analyses were further limited to those pyramidal neurons that we characterized as relatively complete (246; African leopard = 148; cheetah = 98). The examination of dendritic characteristics in only two animals posed statistical challenges, as detailed elsewhere (Johnson et al., 2016). Simple *t* tests and ANOVAs were initially used in an attempt to differentiate species. To evaluate all possible combinations of measures simultaneously with the goal of a summarized qualification of species neuronal differences, a quantitative nonparametric methodology (MARSplines or Multivariate Adaptive Regression Splines; Statistica, release 12; StatSoft, Austin, TX; Friedman, 1991; Hastie, Tibshirani, & Friedman, 2009) was used to evaluate distinctions between the African leopard and the cheetah. This nonparametric approach is robust to violations of normality and nonhomogeneity of variability (for details, see Jacobs et al., 2014; Jacobs, Lee, et al., 2015; Johnson et al., 2016).

The MARSpline analysis was limited to a maximum of 10 basis functions, which enables a regression solution that, unlike a conventional two-dimensional regression that predicts *y* from *x*, can predict in up to 10 dimensions. Third degree interactions were chosen to enable combinations of up to three variables (e.g., Vol, MSL, and DSN) to be used in the underlying equations. The analysis was performed with a penalty of one, where the penalty controls the number of dimensions the analysis can add to solve the problem. Pruning removes relatively less predictive elements of the equations, and a low value for threshold supports the addition of more dimensions. A threshold of 0.0005 and no pruning was used to facilitate a multidimensional analysis to account for both repeated measure and nested effects within species. A Fisher exact test framework was used to calculate a χ^2 statistic to evaluate the null hypothesis of no difference between species. In addition, an R^2 was computed to characterize the amount of differential variability in dendritic measures between species that could be captured. A simple accuracy measure based on correct and incorrect classifications was then calculated.

2.6 | Stereological analysis

Unbiased stereology was used to estimate the length (μm), area (μm^2), and volume (μm^3) of pyramidal neuron somata in M1, V1, and prefrontal cortex, and of gigantopyramidal neuron somata in M1 and V1. This approach resulted in the following six measurements: pyramidal neuron soma length, area, and volume (PSL, PSA, PSV); gigantopyramidal neuron soma length, area, and volume (GSL, GSA, GSV). These parameters were acquired using a Nikon Eclipse 80i microscope equipped with a motorized stage and a color video camera (Qimaging Retiga 2000R), which was coupled to a workstation running StereoInvestigator software (MBF Bioscience, Williston, Vermont; Version 10.50; RRID:SCR-002526). Using criteria from published descriptions (Jacobs et al., 2018; Johnson et al., 2016), regions of interest were identified at low magnification (4X) followed by quantification at high magnification (63X oil objective) using the Nucleator probe. To optimize the sampling and achieve a suitable coefficient of error (CE) below 0.1, pilot studies for length, area, and volume were performed on cresyl violet stained sections for both neuronal types (Dell, Patzke, Spocter, Siegel, & Manger, 2016; Gundersen, 1988). During quantification, the observer (M.A.S.) blindly measured both pyramidal and gigantopyramidal neurons. Separate pilot studies were conducted for each neuronal type in order to optimize the sampling design. The following stereological parameters were applied uniformly across species and cortical region: sampling grid size (350 μm × 350 μm), counting frame size (125 μm × 125 μm), dissector height (10 μm), guard zone (2 μm), and a section interval of 10. After stereological measurements were completed, the data were

TABLE 1 Stereological parameters used for estimating neuron somata dimensions by cortical region in the African lion, African leopard, and cheetah

Species/region	Number of cells counted	Section cut thickness (μm)	Measured mounted thickness (μm)	Number of sections	Number of sampling sites	Average CE
African lion						
Motor	286	50	15.6	6	114	0.07
Visual	222	50	17.3	6	89	0.05
Frontal	266	50	16.3	6	130	0.08
African leopard						
Motor	361	30	20.3	6	122	0.07
Visual	75	30	18.2	6	106	0.05
Frontal	73	30	22.2	6	111	0.09
Cheetah						
Motor	332	50	17.1	6	94	0.04
Visual	187	50	18.5	6	120	0.06
Frontal	336	50	15.6	6	131	0.08

Abbreviation: CE, coefficient of error.

distributed in a bimodal manner using a Finite Mixed model implemented in the R programming language (www.r-project.org/, RRID: SCR_001905), and the gigantopyramidal neurons were quantitatively distinguished from the typical pyramidal neurons using the statistically computed cutoff value generated from the model. The observed ranges of each neuronal type and the number of neurons sampled are provided in Table 1. Once the neuronal cell types were partitioned into the two discrete groups, an ANOVA was used to test for significant group differences in neuronal morphometry, followed by post hoc multiple comparisons (Dunn's post hoc with Bonferroni correction) for cases with significant overall effects. Linear regression was used to investigate scale relationships between brain mass and neuronal cell parameters, and we performed a linear regression analysis (Ordinary Least Squares, OLS) on the logarithmically transformed (base 10) data. To facilitate a broader analysis, additional data from other Carnivora (Jacobs et al., 2018) were included. Finally, to account for the effects of species relatedness, a Phylogenetic Generalized Least Squares (PGLS) analysis was performed in the R "caper" package (Orme et al., 2013) using data based on the mammalian supertree (Bininda-Edmonds et al., 2007, 2008).

3 | RESULTS

3.1 | Overview of Golgi staining

Golgi-stained tissue in the African leopard and cheetah was of relatively high quality, but incomplete impregnation of neurons was observed in the African lion tissue samples (African lion: Figure 2; African leopard: Figure 3; cheetah: Figure 4). Quantitative measures for all neuronal types among the three species are presented in Table 2 for each cortical region, although the data for the lion are clearly underestimations and could not be used for comparative purposes. Overall, a diverse population of neurons was documented across all three species. Qualitative descriptions of neuronal morphology for the lion are illustrated in Figure 5. Spiny and aspiny neuron tracings are depicted for the African leopard in Figures 6 and 7, respectively, and for each cortical region in the cheetah (prefrontal cortex: Figure 8; M1: Figure 9; V1: Figure 10; aspiny interneurons: Figure 11). Spiny neurons included typical pyramidal neurons along with several other variants, each discussed below. Aspiny neurons were further classified into multipolar or bipolar interneurons, as well as neurogliaform cells. Sholl analyses for all neuronal types in the African leopard and cheetah are provided in Figure 12.

3.2 | Spiny neurons

3.2.1 | Typical pyramidal neurons

These neurons were the most frequently traced across all cortical regions and species (African lion: Figure 2a,c, Figure 5e-i,s-w; African leopard: Figure 3a,c, Figure 6a-c,k,q-r; cheetah: Figure 4b,c, Figure 8a–f, Figure 9a–f, Figure 10a–h). Typical pyramidal neurons were characterized by triangular, rounded, or elongated somata (average soma depth range from Table 2 = 738–1,134 μm) and a singular, branching apical dendrite ascending towards the pial surface with several oblique extensions. Their diffuse basilar dendritic skirts contained an average of 4.29 ± 0.07 primary dendrites. In general, most morphological measures in cheetah pyramidal neurons were greater than those in African leopard pyramidal neurons: Vol (by 30%), TDL (by 42%), MSL (by 18%), DSC (by 21%), DSN (by 47%), and soma size (by 41%). Sholl analyses for both species (Figure 12a,j) indicated dendritic intersections peaking near the soma (<200 μm), with denser basilar than apical branching. Dendrites in the cheetah generally extended further from the somata (basilar: ~1,100 μm, apical: ~1,300 μm) than in the African leopard (basilar: ~550 μm, apical: ~950 μm). Inferential analyses of these pyramidal neurons were used to describe cortical regional and interspecies differences (Figure 13), as detailed below in Section 3.4.

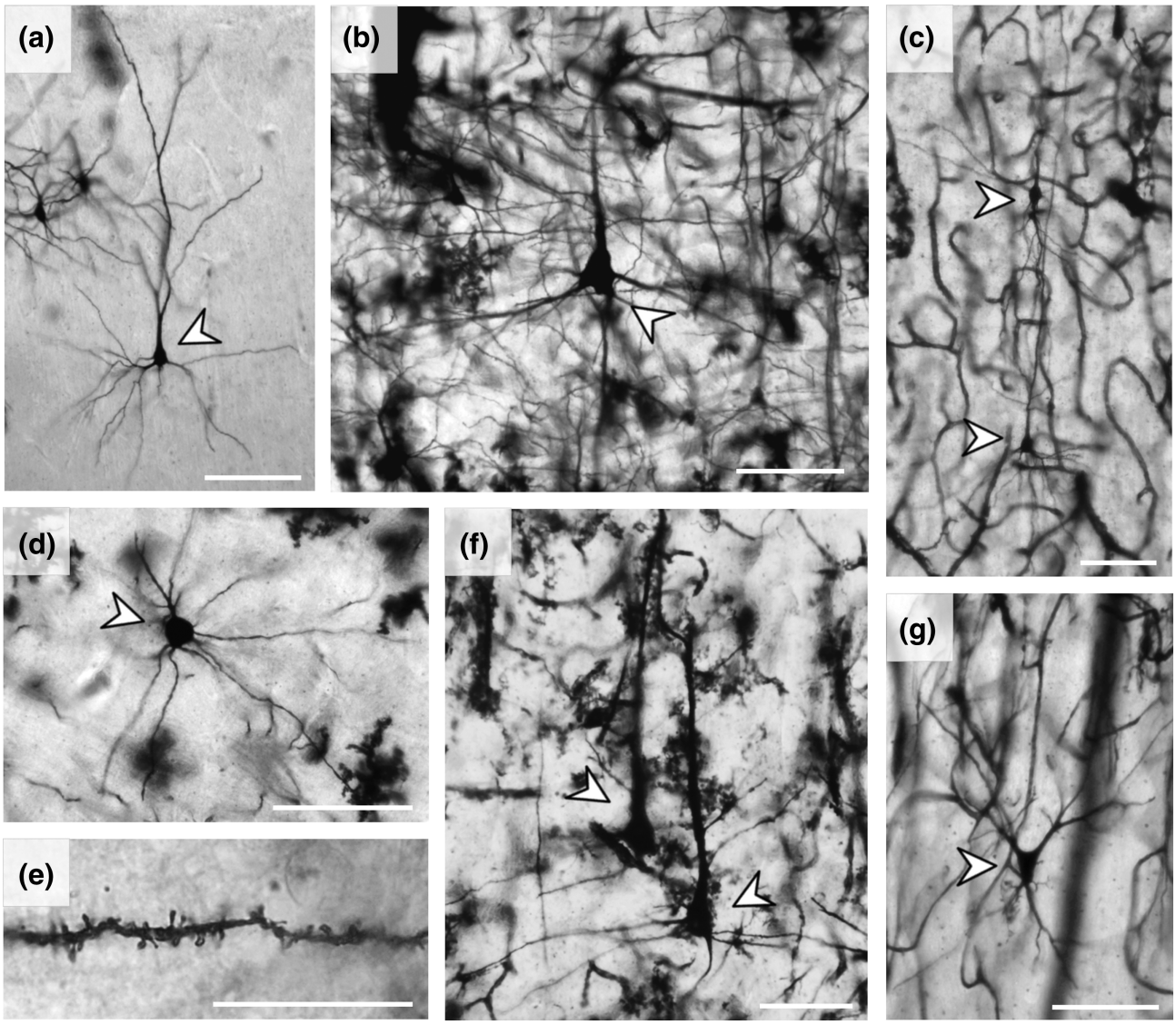

FIGURE 2 Photomicrographs of Golgi-stained neurons in primary visual (a–c) and primary motor (d–g) cortices of the African lion: Typical pyramidal neurons (a, c, see also Figure 5w for lower neuron in c), solitary gigantopyramidal neuron of Meynert (b, see also Figure 5y), aspiny interneuron (d, see also Figure 5p), higher magnification of dendritic spines on the basilar dendrite of a pyramidal neuron (e), primary motor gigantopyramidal neuron (f), and extraverted pyramidal neuron (g). Arrowheads indicate somata. Note the relative incompleteness of dendritic systems in these neurons. Scale bars = 100 μm in a–d, f–g. Scale bar = 50 μm in e

3.2.2 | Extraverted pyramidal neurons

These neurons were traced across all cortical regions and species except for V1 in the African lion and prefrontal cortex in the cheetah (African lion: Figure 2g, Figure 5d,m-n; African leopard: Figure 3d, Figure 6d,l,o–p; cheetah: Figure 4g, Figure 9k–l, Figure 10m–n). Extraverted pyramidal neurons were the most superficial of all neuronal types in the present study, with an average soma depth (as indicated in Table 2) ranging from 463–874 μm. These neurons had an average of 3.28 ± 0.21 primary basilar dendrites per cell (23% fewer than in typical pyramidal neurons) and two distinct apical projections that bifurcated near a round- or triangular-shaped soma. As such, the dendritic extent of these neurons was greater for apical than basilar dendrites. In general, cheetah extraverted pyramidal neurons were larger and more complex than in the African leopard, particularly for TDL (by 55%), DSC (by 39%), and DSN (by 46%), and slightly larger for Vol (by 9%), MSL (by 13%), and soma size (by 9%). Sholl analyses (Figure 12b,k) revealed that the greatest dendritic complexity was near the soma (~100 μm), with apical and basilar branches reaching similar magnitudes of peak intersections in both species.

3.2.3 | M1 gigantopyramidal neurons

These neurons were traced in layer V (average soma depth range from Table 2 = 1,467–1,885 μm) of M1 in all three species (African lion:

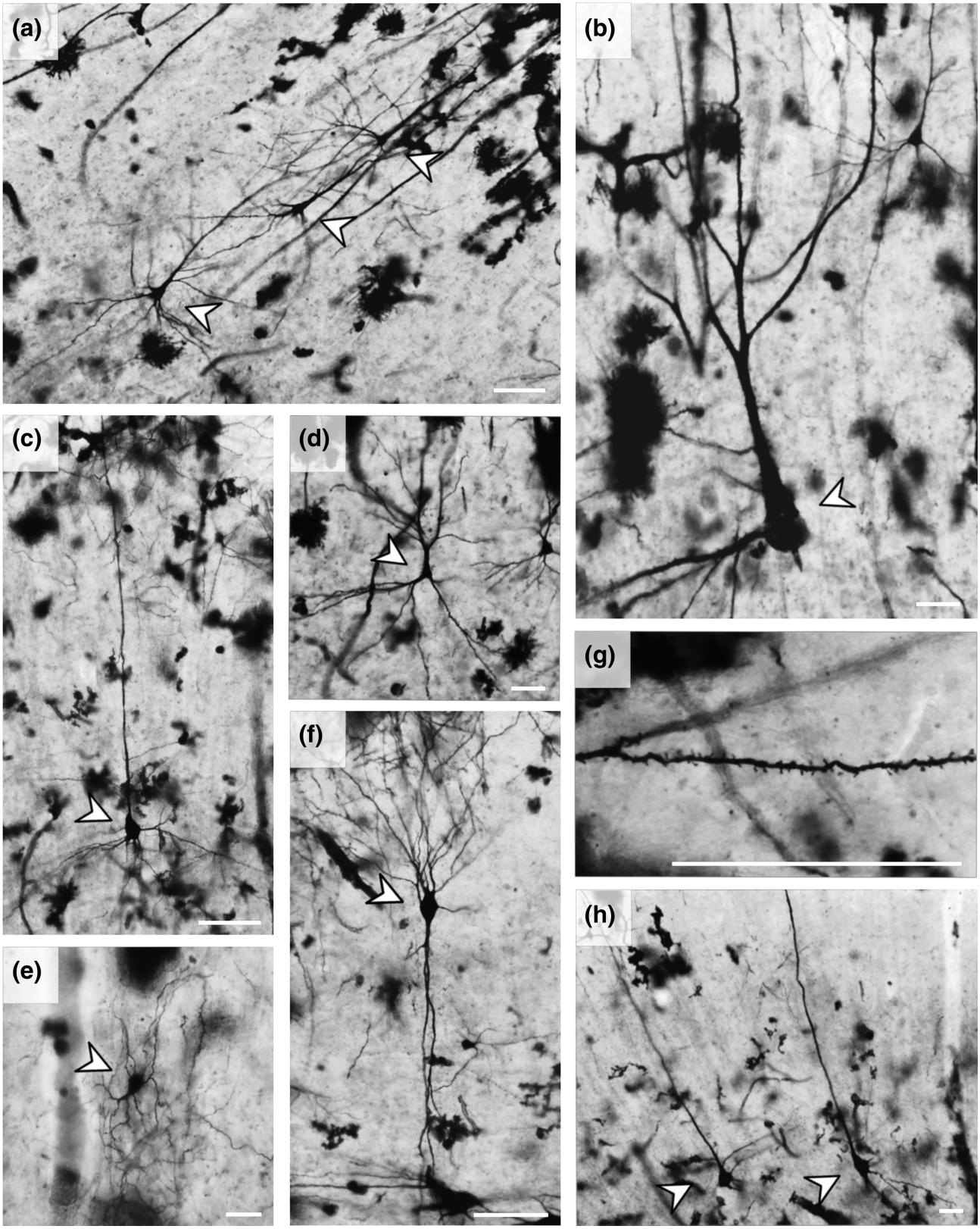

FIGURE 3 Photomicrographs of Golgi-stained neurons in primary motor (a, b, d, g) and primary visual cortices (c, e, f, h) of the African leopard: Typical pyramidal neurons (a, c), primary motor gigantopyramidal neuron (b), extraverted pyramidal neuron (d), neurogliaform neuron (e), aspiny interneuron (f, see also Figure 7m), higher magnification of dendritic spines on the basilar dendrite of a pyramidal neuron (g), and solitary gigantopyramidal neuron of Meynert (h, see also Figure 6s,t). Arrowheads indicate somata. Note the vertical arrangement of the apical dendrites in a. Scale bars = 100 μm

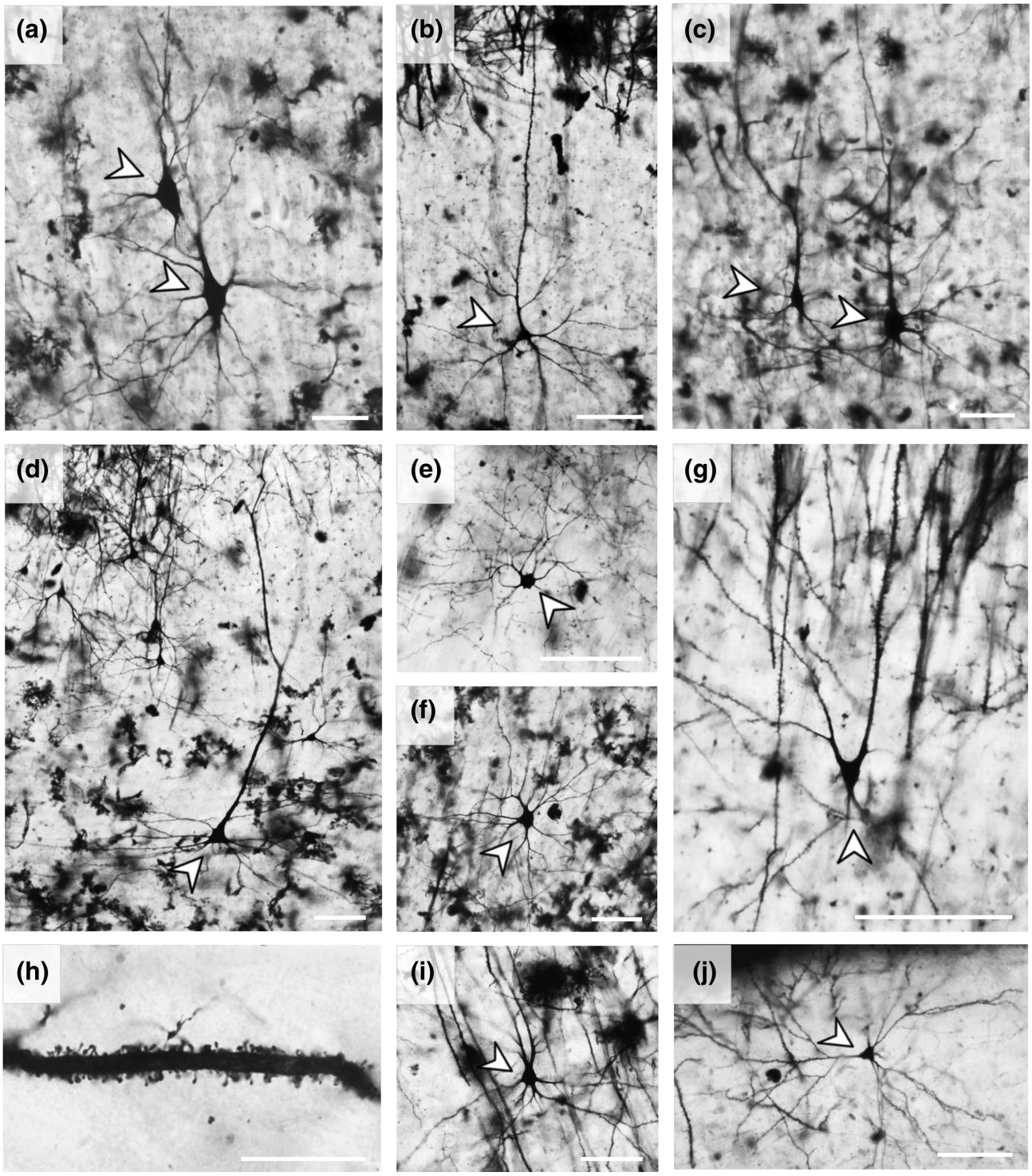

FIGURE 4 Photomicrographs of Golgi-stained neurons in primary motor (a, c, g, i, j), primary visual (b, d, f), and prefrontal (e, h) cortices of the cheetah: Primary motor gigantopyramidal neurons (a, right neuron in c; see also Figure 9g for lower neuron in a; see also Figure 9h for right neuron in c), typical pyramidal neurons (b, left neuron in c; see also Figure 10b for b), solitary gigantopyramidal neuron of Meynert (d), neurogliaform neuron (e), aspiny interneuron (f, see also Figure 11l), extraverted pyramidal neuron (g, see also Figure 9l), higher magnification of dendritic spines on the apical dendrite of a pyramidal neuron (h), multiapical pyramidal neuron (i, see also Figure 9n), and horizontal pyramidal neuron (j, see also Figure 9p). Arrowheads indicate somata. Scale bars = 100 μm in a–g, i, j. Scale bar = 25 μm in h

Figure 2f, Figure 5j–l; African leopard: Figure 3b, Figure 6i,j; cheetah: Figure 4a,c, Figure 9g–j). M1 gigantopyramidal neurons exhibited extremely large somata, with an average cross-sectional area that was 541% greater than in typical pyramidal neurons across all cortical regions, and a thick, ascending apical dendrite. These were the most morphologically varied and idiosyncratic of the spiny neurons. They

TABLE 2 Summary statistics (mean ± *SEM*) for each cortical neuron type quantified in the African lion, African leopard, and cheetah

Species/region/neuron type	n[c]	Vol[d]	TDL[e]	MSL[e]	DSC[f]	DSN[g]	DSD[h]	Soma size[i]	Soma depth[j]
African lion[a]									
Frontal									
Aspiny	2	7,621 ± 2,953	2,866 ± 386	79 ± 33	47 ± 25	–	–	456 ± 8	755 ± 373
Extraverted	3	8,877 ± 486	3,055 ± 346	54 ± 7	59 ± 13	465 ± 1	0.16 ± 0.02	313 ± 22	582 ± 97
Multiapical	1	15,387	3,531	86	41	982	0.28	484	830
Pyramidal	14	11,019 ± 966	3,037 ± 223	70 ± 4	46 ± 4	711 ± 85	0.23 ± 0.02	385 ± 24	738 ± 63
Motor									
Aspiny	5	6,224 ± 817	2,492 ± 192	70 ± 7	36 ± 3	–	–	401 ± 51	989 ± 190
Extraverted	11	10,324 ± 1,089	3,487 ± 414	59 ± 3	59 ± 6	645 ± 183	0.16 ± 0.03	337 ± 28	512 ± 25
Gigantopyramidal	16	49,253 ± 4,580	1,872 ± 215	53 ± 3	35 ± 3	210 ± 36	0.10 ± 0.01	1,771 ± 176	1,885 ± 34
Multiapical	2	13,812 ± 3,086	3,292 ± 278	69 ± 4	48 ± 2	664 ± 75	0.20 ± 0.01	419 ± 15	1,106 ± 387
Pyramidal	32	15,199 ± 1,035	4,135 ± 187	68 ± 2	61 ± 3	722 ± 64	0.17 ± 0.01	399 ± 15	984 ± 87
Visual									
Aspiny	9	3,544 ± 688	2,163 ± 359	60 ± 6	36 ± 4	–	–	329 ± 41	984 ± 107
Inverted	3	8,906 ± 1,101	3,302 ± 302	74 ± 7	46 ± 9	533 ± 250	0.17 ± 0.09	374 ± 46	1,747 ± 428
Meynert[b]	6	33,878 ± 4,984	3,813 ± 808	74 ± 10	48 ± 6	332 ± 156	0.07 ± 0.02	1,655 ± 125	1,736 ± 91
Pyramidal	58	12,493 ± 1,283	3,672 ± 162	61 ± 2	61 ± 2	777 ± 58	0.20 ± 0.01	372 ± 47	1,087 ± 69
African leopard									
Frontal									
Aspiny	6	3,736 ± 504	2,261 ± 360	63 ± 9	37 ± 5	–	–	225 ± 18	1,211 ± 118
Extraverted	3	7,185 ± 755	3,349 ± 645	68 ± 12	50 ± 6	1,388 ± 521	0.39 ± 0.08	267 ± 45	874 ± 131
Inverted	4	6,843 ± 2,029	2,161 ± 369	60 ± 4	36 ± 6	568 ± 179	0.24 ± 0.06	321 ± 58	1,484 ± 305
Multiapical	1	30,469	2,955	70	42	2,421	0.82	584	1,473
Neurogliaform	1	1,125	1,204	23	52	–	–	151	1,087
Pyramidal	20	8,809 ± 716	3,337 ± 172	64 ± 4	53 ± 3	1,703 ± 169	0.50 ± 0.04	284 ± 14	1,134 ± 106
Motor									
Aspiny	33	8,512 ± 565	2,899 ± 136	77 ± 3	39 ± 2	-	-	444 ± 25	957 ± 44
Extraverted	9	8,317 ± 1,559	2,759 ± 272	66 ± 6	42 ± 2	782 ± 109	0.28 ± 0.03	235 ± 20	614 ± 62
Gigantopyramidal	17	67,235 ± 5,026	3,001 ± 389	130 ± 8	25 ± 3	595 ± 95	0.20 ± 0.01	2,548 ± 158	1,467 ± 40
Horizontal	2	6,537 ± 2,640	2,744 ± 785	57 ± 8	48 ± 8	524 ± 221	0.18 ± 0.03	337 ± 121	1,429 ± 168
Inverted	3	8,085 ± 556	2,550 ± 308	81 ± 11	33 ± 5	504 ± 235	0.20 ± 0.10	344 ± 82	1,407 ± 228
Pyramidal	87	10,482 ± 482	3,379 ± 89	65 ± 1	53 ± 1	846 ± 36	0.25 ± 0.01	271 ± 10	829 ± 33
Visual									
Aspiny	13	6,239 ± 1,588	3,074 ± 299	67 ± 4	47 ± 5	–	–	305 ± 54	910 ± 81
Extraverted	5	4,675 ± 791	2,633 ± 440	52 ± 3	50 ± 7	1,105 ± 339	0.40 ± 0.10	202 ± 33	463 ± 15
Horizontal	2	4,567 ± 299	3,282 ± 179	67 ± 8	49 ± 3	1,020 ± 116	0.31 ± 0.05	220 ± 14	1,152 ± 188
Inverted	1	5,672	3,451	65	53	1,389	0.40	132	913
Meynert[b]	4	22,944 ± 1,565	2,898 ± 180	97 ± 10	31 ± 5	314 ± 87	0.11 ± 0.03	1,037 ± 85	1,373 ± 20
Neurogliaform	4	5,051 ± 1,795	3,475 ± 512	53 ± 7	66 ± 3	–	–	216 ± 61	725 ± 167
Pyramidal	42	9,076 ± 1,001	3,430 ± 172	64 ± 2	54 ± 2	1,104 ± 73	0.33 ± 0.02	277 ± 25	921 ± 59
Cheetah									
Frontal									
Aspiny	9	4,735 ± 879	2,884 ± 423	89 ± 6	35 ± 7	–	–	250 ± 14	882 ± 152
Horizontal	3	13,131 ± 1,348	5,106 ± 308	80 ± 4	65 ± 7	2,315 ± 398	0.45 ± 0.06	418 ± 53	1,140 ± 426
Inverted	1	26,434	8,596	85	101	2,380	0.28	542	1,804
Multiapical	3	10,915 ± 4,134	4,230 ± 1,199	78 ± 11	53 ± 10	1,535 ± 571	0.35 ± 0.05	462 ± 110	1,246 ± 349

(Continues)

TABLE 2 (Continued)

Species/region/neuron type	n[c]	Vol[d]	TDL[e]	MSL[e]	DSC[f]	DSN[g]	DSD[h]	Soma size[i]	Soma depth[j]
Neurogliaform	6	4,019 ± 1,224	5,514 ± 1,217	58 ± 2	94 ± 18	-	-	182 ± 22	896 ± 37
Pyramidal	29	14,750 ± 1,774	5,708 ± 268	90 ± 3	64 ± 2	2,561 ± 179	0.44 ± 0.02	394 ± 22	1,093 ± 76
Motor									
Aspiny	9	8,392 ± 1,675	3,240 ± 265	111 ± 10	30 ± 3	–	–	465 ± 63	925 ± 43
Extraverted	6	6,616 ± 1,067	4,215 ± 292	72 ± 4	59 ± 4	1,579 ± 224	0.37 ± 0.03	225 ± 21	530 ± 72
Gigantopyramidal	29	51,059 ± 3,965	4,827 ± 365	90 ± 3	55 ± 4	411 ± 73	0.08 ± 0.01	2,210 ± 163	1,637 ± 29
Horizontal	1	5,764	5,077	72	71	1,972	0.39	228	227
Inverted	2	9,379 ± 4,356	5,289 ± 523	97 ± 19	58 ± 17	1,720 ± 192	0.33 ± 0.07	365 ± 14	1,342 ± 246
Multiapical	9	27,719 ± 4,952	5,912 ± 431	81 ± 4	73 ± 4	1,782 ± 118	0.31 ± 0.02	592 ± 67	1,473 ± 106
Pyramidal	54	12,069 ± 1,413	4,350 ± 154	77 ± 2	57 ± 2	1,172 ± 62	0.28 ± 0.02	365 ± 36	843 ± 54
Visual									
Aspiny	15	6,891 ± 1,087	3,170 ± 239	78 ± 4	42 ± 3	–	–	432 ± 50	1,106 ± 80
Extraverted	2	10,856 ± 4,181	4,826 ± 490	62 ± 2	78 ± 11	1,010 ± 315	0.22 ± 0.09	336 ± 116	484 ± 105
Meynert[b]	11	27,099 ± 2,456	5,963 ± 478	94 ± 8	66 ± 8	742 ± 113	0.14 ± 0.03	940 ± 105	1,388 ± 86
Pyramidal	44	12,471 ± 877	4,820 ± 144	67 ± 1	73 ± 2	1,267 ± 92	0.26 ± 0.01	411 ± 52	784 ± 39

[a]Quantitative measures for African lion neurons are underestimations, especially for spine measures (DSN and DSD), due to the relatively incomplete Golgi impregnation.
[b]Meynert = solitary gigantopyramidal neuron of Meynert.
[c]Number of neurons traced.
[d]Volume in μm^3.
[e]Length in μm (TDL, total dendritic length; MSL, mean segment length).
[f]Number of segments per neuron (DSC, number of dendritic segments).
[g]Number of spines per neuron (DSN, total number of spines on dendritic segments).
[h]Number of spines per μm of dendritic length (DSD, average number of spines per μm of dendritic length).
[i]Soma size in μm^2.
[j]Soma depth in μm from the pial surface.

were characterized by a dense basilar skirt (average number of primary basilar dendrites = 5.76 ± 0.21) that predominantly descended, but sometimes projected in all directions from the soma. Some of these neurons possessed a single descending taproot dendrite. M1 gigantopyramidal neurons in the African leopard were generally larger in dendritic Vol (by 32%), MSL (by 45%), and soma size (by 15%), and were more spinous (DSN, by 45%; DSD, by 145%), than in the cheetah. However, cheetah neurons showed a greater TDL (by 61%) and DSC (by 122%) than African leopard neurons. Sholl analyses (Figure 12c,l) further supported the finding of greater dendritic branching in cheetah neurons than in African leopard neurons, with higher peak intersections near the soma (<200 μm). In both species, these neurons demonstrated the furthest dendritic extent of all neuron types (~1,350 to 1,400 μm), although these values were still attenuated due to truncation of dendritic arbors from sectioning.

3.2.4 | Solitary gigantopyramidal neurons of Meynert

These neurons were observed in layers III-V (average soma depth range from Table 2 = 1,373–1,736 μm) of V1 across all species (African lion: Figure 2b, Figure 5x–z; African leopard: Figure 3h, Figure 6s–t; cheetah: Figure 4d, Figure 10i–l). As with M1 gigantopyramidal neurons, these neurons had a large soma and a thick, ascending apical dendrite. Their often laterally projecting, dense basilar skirts averaged 7.05 ± 0.58 primary dendrites, the highest among all neuronal types. In contrast, average DSD ranged from 0.07 to 0.14 (Table 2), the lowest of all neuronal types. Soma size was greater than that of typical pyramidal neurons (by 241%) and smaller than that of M1 gigantopyramidal neurons (by 47%). In general, solitary gigantopyramidal neurons of Meynert in the cheetah exhibited greater dendritic/spine complexity than those in the African leopard, as measured by Vol (by 18%), TDL (by 106%), DSC (by 113%), DSN (by 136%), and DSD (by 24%). However, in the African leopard, soma size was larger (by 10%) and MSL was similar when compared to the cheetah. For both species, Sholl analyses (Figure 12d,m) showed similar peak intersections near the soma (<200 μm) and dendritic branches reaching ~1,350 μm. As with M1 gigantopyramidal neurons, these values were attenuated due to incompleteness of tracings from sectioning.

3.2.5 | Multiapical pyramidal neurons

These neurons were traced in the prefrontal cortex and M1 of the African lion (Figure 5c) and cheetah (Figure 4i, Figure 8k,l, Figure 9n, o), as well as in the prefrontal cortex of the African leopard

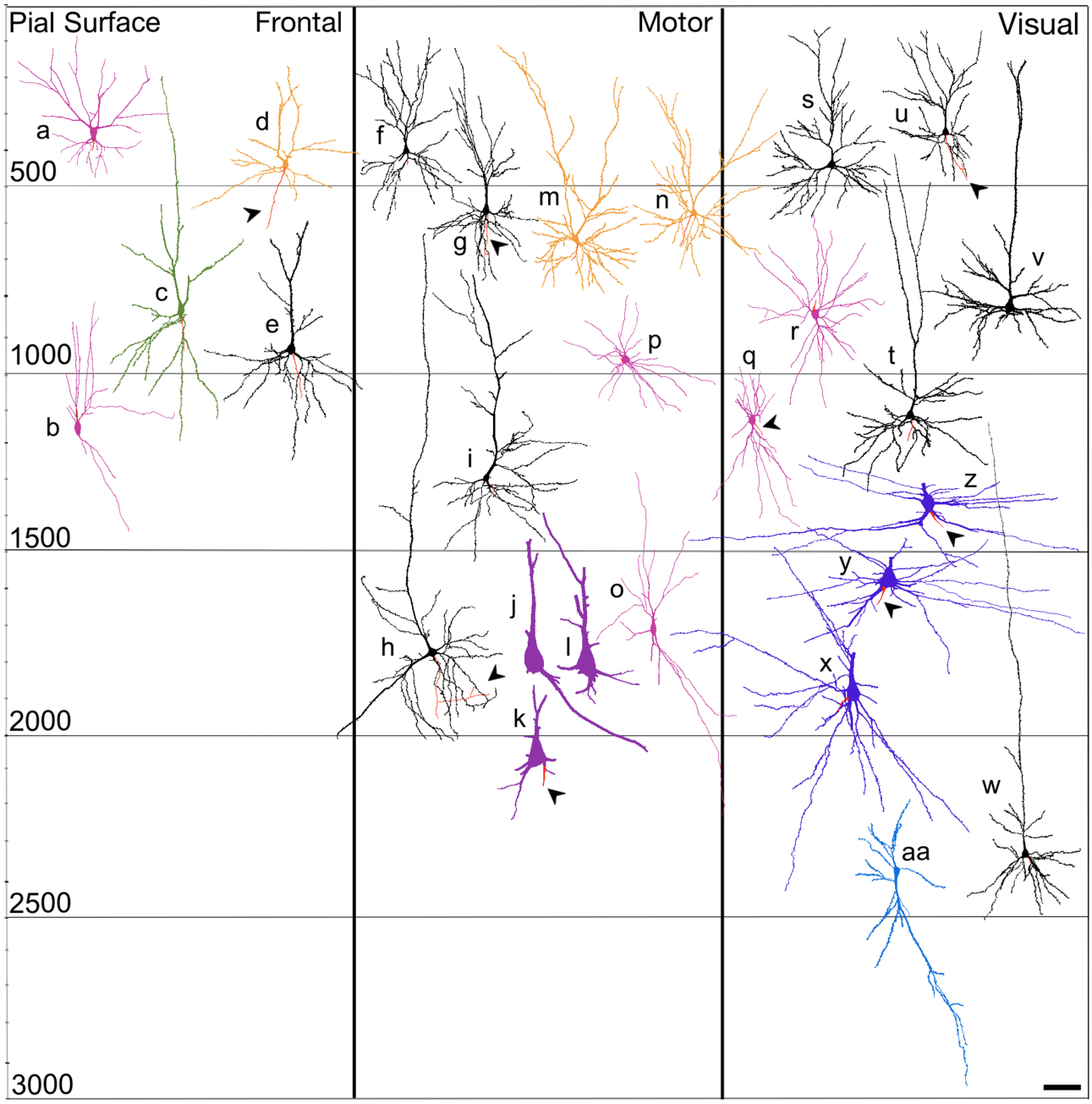

FIGURE 5 Neurolucida tracings of spiny and aspiny neurons in the prefrontal (a–e), primary motor (f–p), and primary visual (q–aa) cortices of the African lion indicating relative soma depth from the pial surface (in μm): Aspiny interneurons (a,b,o–r), multiapical pyramidal neuron (c), extraverted pyramidal neurons (d,m,n), typical pyramidal neurons (e–i,s–w), primary motor gigantopyramidal neurons (j–l), solitary gigantopyramidal neurons of Meynert (x–z), and inverted pyramidal neuron (aa). Arrowheads indicate axons. Scale bar = 100 μm [Color figure can be viewed at wileyonlinelibrary.com]

(Figure 6e). They were characterized by several apical projections ascending towards the pial surface and by extensive basilar skirts (average number of primary dendrites = 4.69 ± 0.36) similar to those observed in typical pyramidal neurons. Multiapical pyramidal neurons were generally deep in the cortex, with an average soma depth range of 830–1,473 μm (Table 2). Average DSD ranged from 0.20 to 0.85 (Table 2), the highest among spiny neurons. Sholl analyses (Figure 12e,n) demonstrated patterns similar to those observed in other neuronal types. However, all measurements were highly variable due to the limited number of neurons traced. Because only one multiapical neuron was traced in the African leopard, comparisons of measures between species were not warranted.

3.2.6 | Inverted pyramidal neurons

These neurons were traced in V1 of the African lion (Figure 5aa), all cortical regions of the African leopard (Figure 6f–h,u), and in

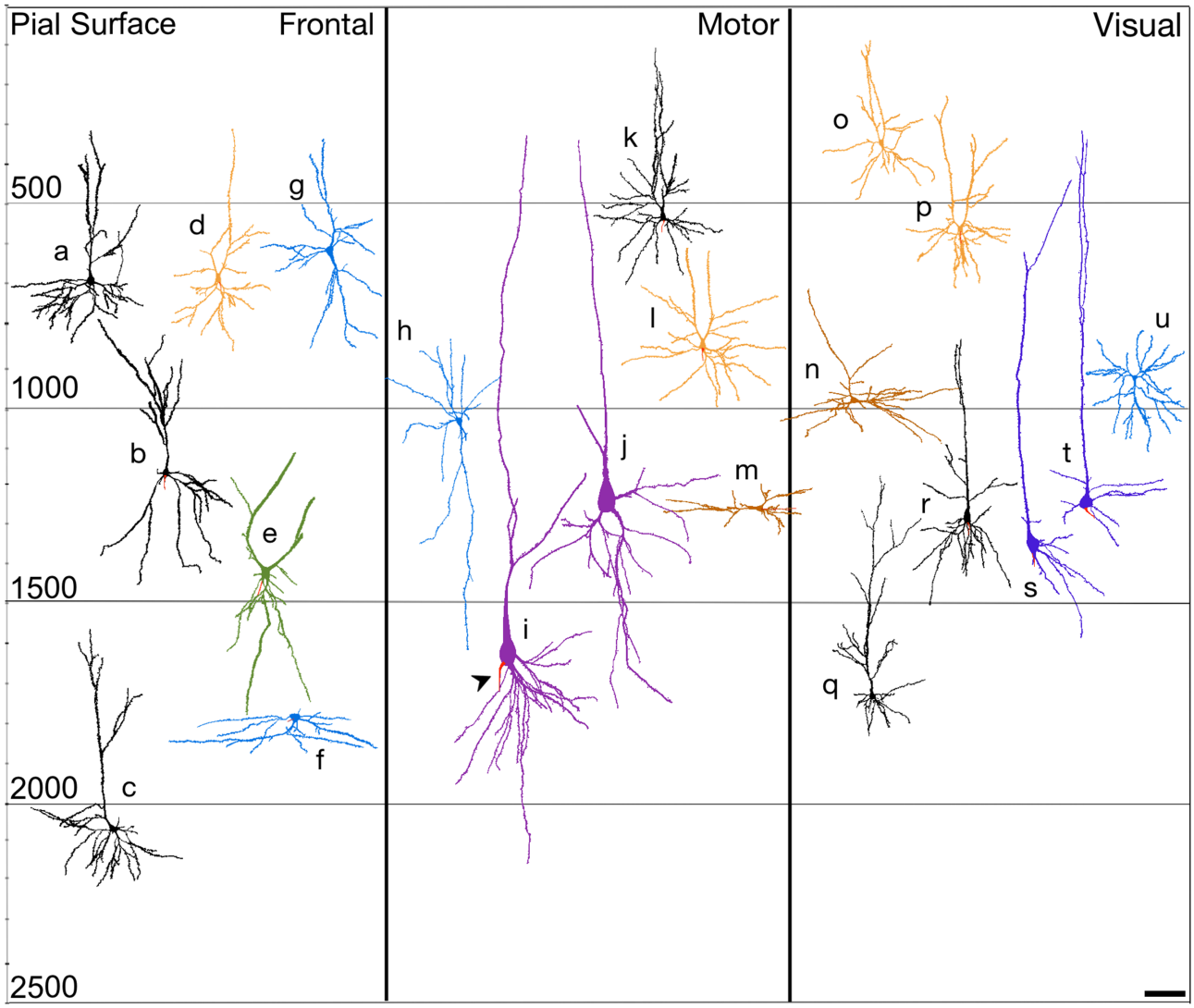

FIGURE 6 Neurolucida tracings of spiny neurons in the prefrontal (a–g), primary motor (h–m), and primary visual (n–u) cortices of the African leopard indicating relative soma depth from the pial surface (in μm): Typical pyramidal neurons (a–c, k, q–r), extraverted pyramidal neurons (d, l, o–p), multiapical pyramidal neuron (e), inverted pyramidal neurons (f–h, u), primary motor gigantopyramidal neurons (i, j), horizontal pyramidal neurons (m, n), and solitary gigantopyramidal neurons of Meynert (s, t). Arrowhead indicates axon. Scale bar = 100 μm [Color figure can be viewed at wileyonlinelibrary.com]

prefrontal cortex and M1 of the cheetah (Figure 8g, Figure 9m). Similar to typical pyramidal neurons, inverted pyramidal neurons had triangular-shaped somata, but were oriented differently, with descending apical dendrites and ascending basilar skirts. These neurons were also some of the deepest neurons sampled, with average soma depth ranging from 913 to 1,804 μm (Table 2). The average number of primary basilar dendrites was 4.29 ± 0.53. Inverted neurons in the cheetah were generally larger (Vol, by 110%; TDL, by 159%; MSL, by 36%; DSC, by 95%; soma size, by 39%) and more spinous (DSN, by 200%; DSD, by 28%) than in the African leopard. Sholl analyses (Figure 12f,o) revealed intersections peaking and diminishing closer to the soma (~450 to 650 μm) than in other neuronal types. Sizeable differences in dendritic arborization between the African leopard and cheetah were observed, although the small numbers of tracings in the cheetah make quantitative comparisons problematic.

3.2.7 | Horizontal pyramidal neurons

These neurons were traced in the African leopard M1 and V1 (Figure 6m,n) and cheetah prefrontal cortex and M1 (Figure 4j, Figure 8h–j, Figure 9p). Horizontal neurons were classified by their small, oblong somata (average soma depth range from Table 2 = 227–1,140 μm) with apical dendrites projecting laterally from opposite poles. Dendritic skirts contained on average 3.75 ± 0.37 primary dendrites. These neurons tended to be larger in the cheetah (Vol, by 103%; TDL, by 69%; MSL, by 25%; DSC, by 37%; soma size, by 33%) and spinier (DSN, by 189%; DSD, by 75%) than in the African leopard. Sholl analyses (Figure 12g,p) revealed a high number of intersections close to the soma (~100 μm) and dendritic branches ending closest to the soma of all neuronal types (~350 to 600 μm). As with other neuron types, horizontal pyramidal neurons in the cheetah exhibited a

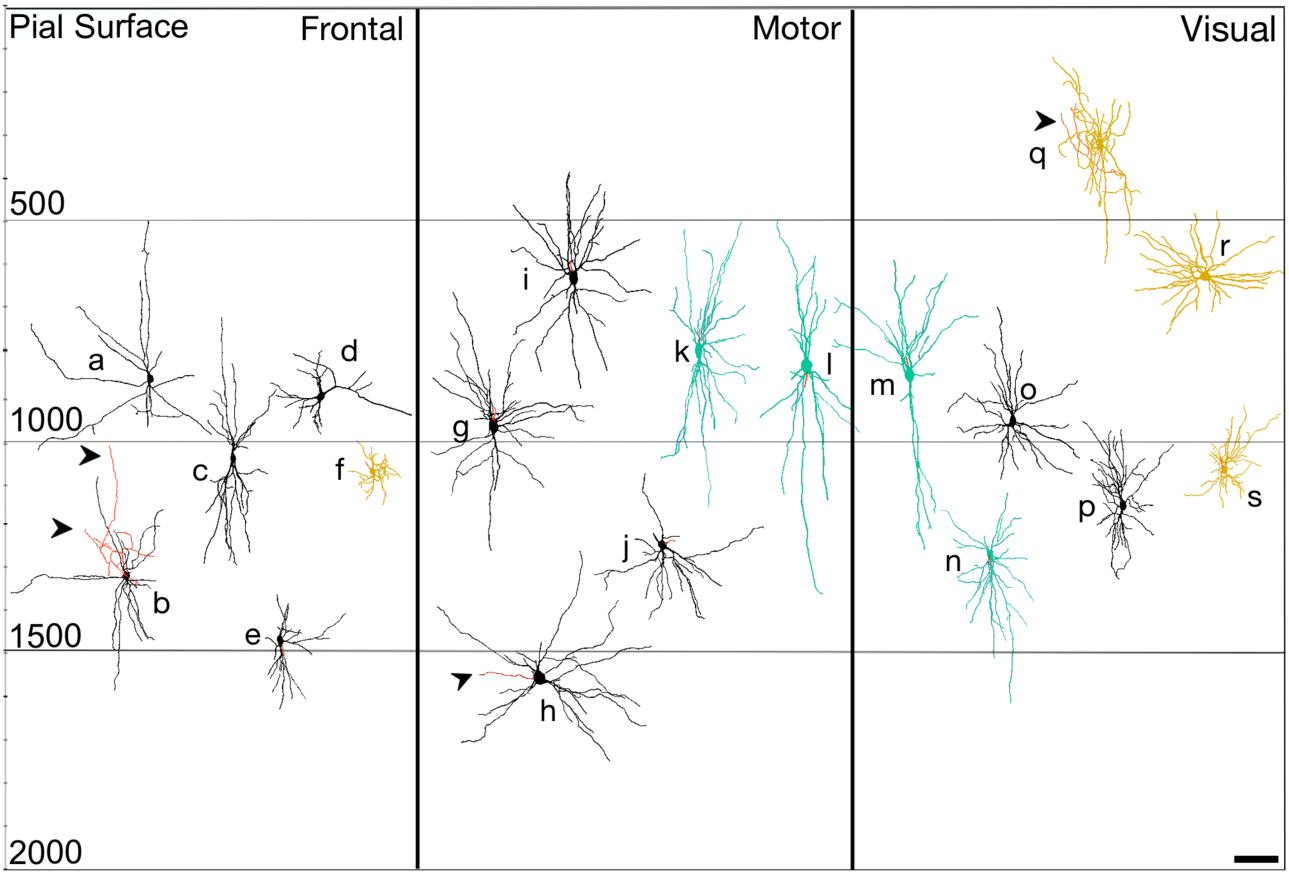

FIGURE 7 Neurolucida tracings of aspiny interneurons in the prefrontal (a–f), primary motor (g–l), and primary visual (m–s) cortices of the African leopard indicating relative soma depth from the pial surface (in μm): Aspiny multipolar (a–e, g–j, o–p), neurogliaform (f, q–s), and aspiny bipolar (k–n) neurons. Arrowheads indicate axons. Scale bar = 100 μm [Color figure can be viewed at wileyonlinelibrary.com]

greater number of branch intersections than those in the African leopard.

3.3 | Aspiny neurons

3.3.1 | Aspiny interneurons

These neurons were traced across all cortical regions in the three species (African lion: Figure 2d, Figure 5a,b,o–r; African leopard: Figure 3f, Figure 7a–e,g–p; cheetah: Figure 4f, Figure 11a,d–o). They averaged 5.50 ± 0.16 primary dendritic branches, and soma depth range averaged between 614 and 1,211 μm (Table 2). These neurons were further divided into two subtypes: (a) multipolar neurons (Figure 7a–e,g–j,o–p, Figure 11a,e–g,k–o), characterized by diffuse dendritic projections from the rounded soma, and (b) bipolar neurons (Figure 7k–n, Figure 11d,h–j), classified by two distinct dendritic bundles vertically projecting from opposite poles of the soma. In general, aspiny neurons in the cheetah were slightly larger than in the African leopard neurons in terms of TDL (by 8%) and MSL (by 23%), but smaller in terms of Vol (by 9%) and DSC (by 9%), with comparable soma sizes. Sholl analyses (Figure 12h,q) demonstrated comparable dendritic branching between species extending to ~600 μm.

3.3.2 | Neurogliaform cells

These neurons were traced in the African leopard prefrontal cortex and V1 (Figure 3e, Figure 7f,q–s) and cheetah prefrontal cortex (Figure 4e, Figure 11b,c). Neurogliaform neurons were characterized by relatively small, circular somata possessing an extremely dense dendritic plexus of thin branches projecting in all directions. Average soma depth ranged from 725 to 1,087 μm (Table 2), and neurons contained an average of 6.36 ± 0.90 primary dendritic segments. Neurogliaform cells in the African leopard were slightly larger than in the cheetah for dendritic Vol (by 6%) and soma size (by 11%), but these neurons in the cheetah were more dendritically complex than in the African leopard for TDL (by 83%), MSL (by 23%), and DSC (by 50%). Sholl analyses (Figure 12i,r) further indicated greater dendritic complexity in neurons of the cheetah than in the African leopard with a higher magnitude of intersections near the soma (~100 μm). With respect to peak intersections, neurogliaform cells were the most complex of all traced neuronal types.

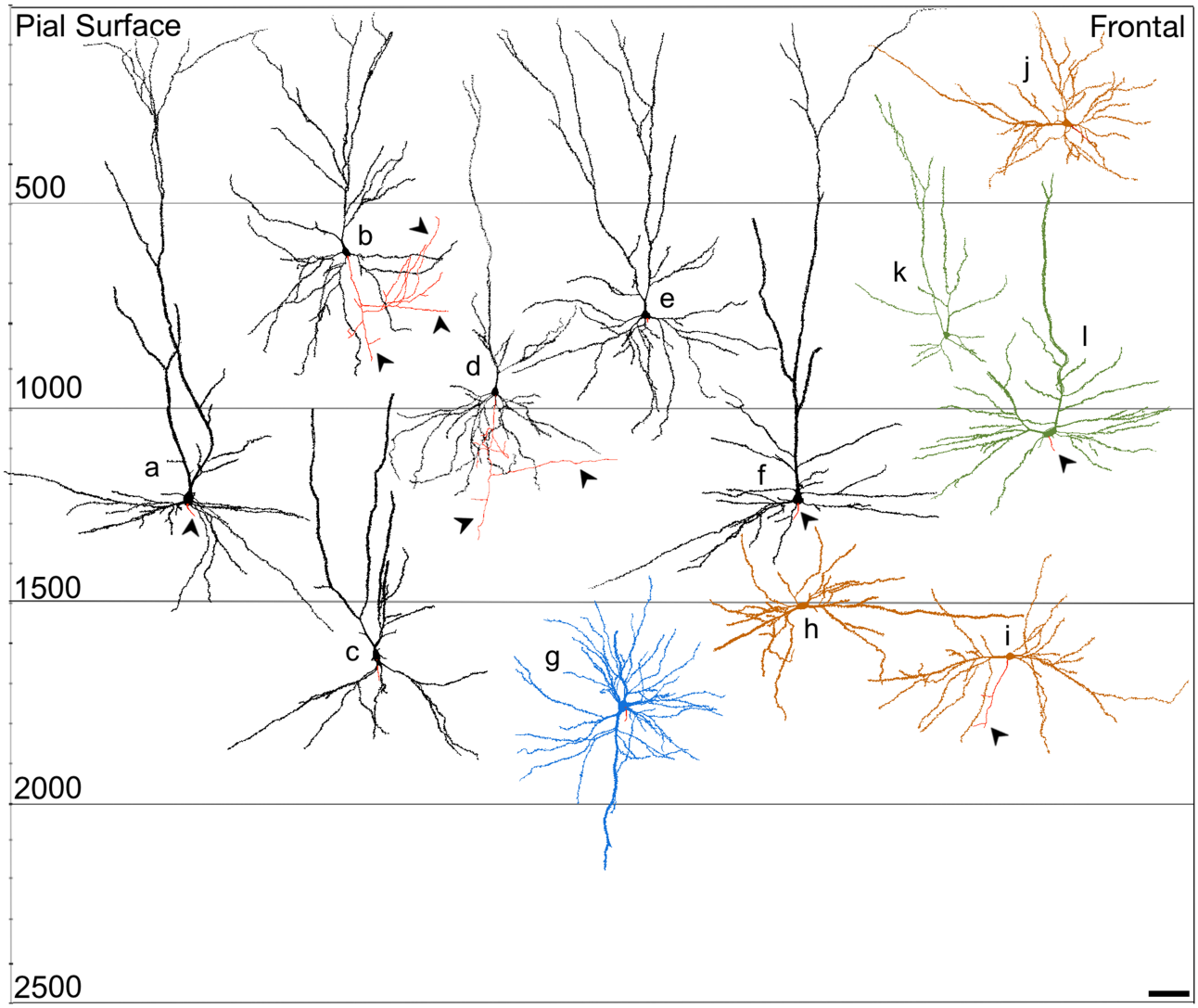

FIGURE 8 Neurolucida tracings of spiny neurons in the prefrontal cortex of the cheetah indicating relative soma depth from the pial surface (in μm): Typical pyramidal (a–f), inverted pyramidal (g), horizontal pyramidal (h–j), and multiapical pyramidal (k–l) neurons. Arrowheads indicate axons. Note the greater dendritic complexity and size of the typical pyramidal neurons (a–f) in the frontal region compared to those in the other cortical regions for the cheetah (see Figures 9 and 10). Scale bar = 100 μm [Color figure can be viewed at wileyonlinelibrary.com]

3.4 | Regional and interspecies dendritic variation for pyramidal neurons

Only typical pyramidal neurons, the most numerously traced neuronal type, were used to examine variation in dendritic and spine measures across cortical regions. For regional differences, a total of 276 pyramidal neurons were included (African leopard = 149; cheetah = 127). Descriptive results for both species across prefrontal cortex, M1, and V1 are provided in Figure 13. Although pyramidal neurons in the African leopard showed minimal regional differences in all measures except for DSD and DSN, prefrontal pyramidal neurons in the cheetah appeared to have longer and spinier dendrites than in the other two cortical regions for all measures.

Results from the MARSplines analysis rejected the null hypothesis of no differentiation in dendritic measures between species, $\chi^2_{(1)} = 133.83$, $p \leq .0001$. This suggested that typical pyramidal neurons were consistently more complex in the cheetah than in the African leopard (Figure 13). This procedure also provided a qualitative assessment of the relative importance of each predictor (i.e., Vol, TDL, etc.) in identifying if a neuron belonged or did not belong to a particular species. Importance was measured by counting the number of times each neuronal variable was entered into the regression equations. The relative importance of each dendritic measure in the MARSplines analysis was as follows: Vol = 5; MSL = 4; TDL = 3; DSC, DSN and DSD = 0, suggesting that Vol and MSL were particularly important predictors of dendritic differences between species for the two animals examined. Only 31 out of 246 neurons were misclassified, resulting in an 87.4% correct differentiation. The R^2 was 0.544, indicating that over 54% of the differential variability in neuronal structure was accounted for between felids.

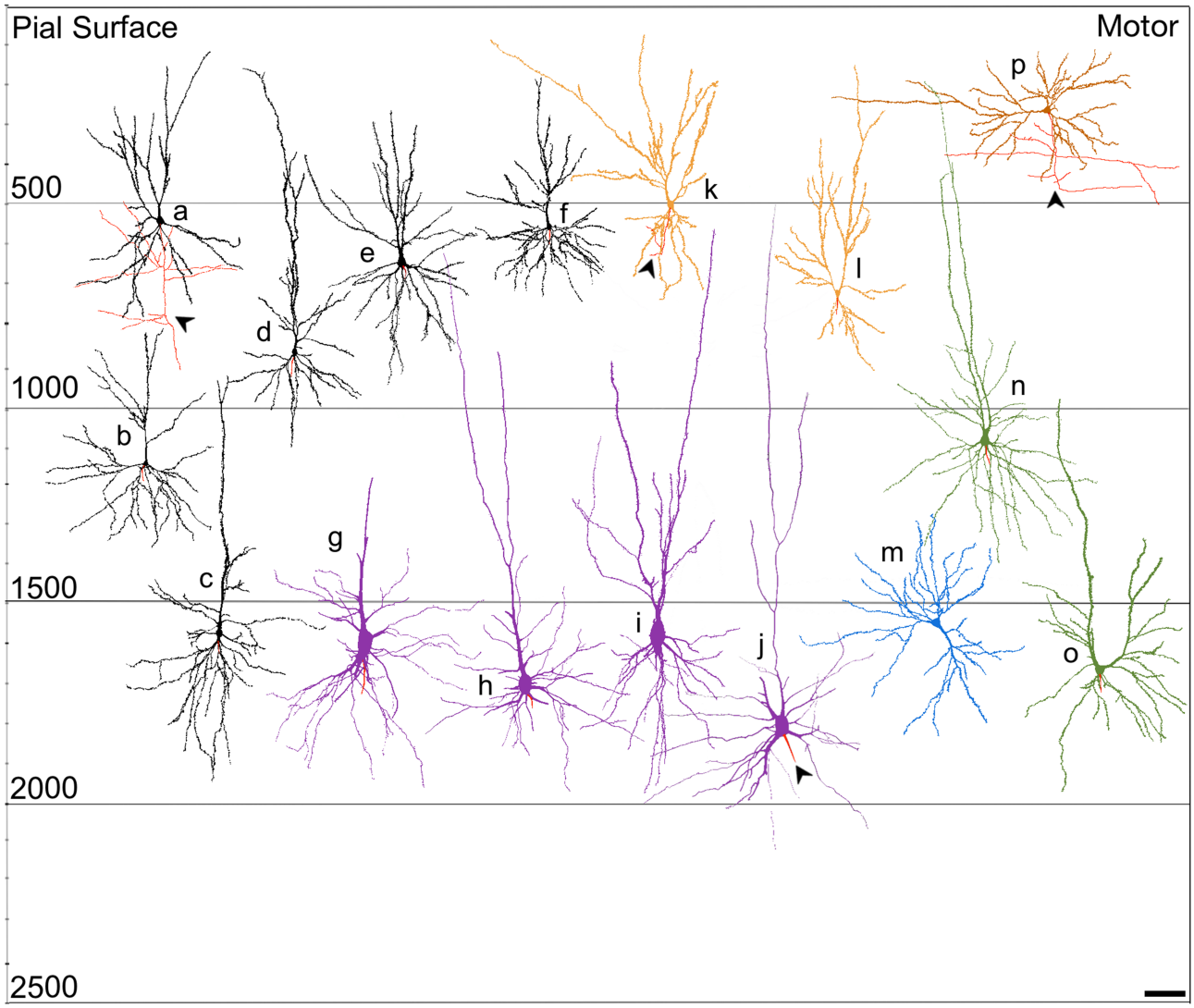

FIGURE 9 Neurolucida tracings of spiny neurons in the primary motor cortex of the cheetah indicating relative soma depth from the pial surface (in μm): Typical pyramidal (a–f), primary motor gigantopyramidal (g–j), extraverted pyramidal (k–l), inverted pyramidal (m), multiapical pyramidal (n,o), and horizontal pyramidal (p) neurons. Arrowheads indicate axons. Note the relatively large soma size of the primary motor gigantopyramidal neurons (g–j) in layer V. Scale bar = 100 μm [Color figure can be viewed at wileyonlinelibrary.com]

3.5 | Stereological analysis of neuronal somata sizes

Cytoarchitectural characteristics of the three isocortical regions for each felid are provided in Figure 14, which illustrates an agranular prefrontal cortex, and an M1 characterized by a thin layer IV with the presence of large gigantopyramidal neuron somata in layer V. For all three species, mini-columns were evident in the deeper layers of V1. The six soma measures (PSL, PSA, PSV; GSL, GSA, GSV) examined in the stereological analysis are provided in Figure 15. For typical pyramidal neurons, soma measures across the three species were similar within each cortical region. The greatest variability was exhibited by PSA (Figure 15b) in the frontal cortex. Soma measures varied more in gigantopyramidal neurons than in pyramidal neurons. For gigantopyramidal neurons, soma measures tended to be much higher in M1 than in V1, especially for GSA (Figure 15e) and GSV (Figure 15f). All soma measures for gigantopyramidal neurons were similar between the lion and the cheetah, both being greater than in the African leopard.

Using an ANOVA, we tested for species level differences in mean soma parameters. In M1, no significant species differences were found for PSV, Welch $F_{(2,\ 976)} = 2.66$, $p \leq .07$, and PSA, Welch $F_{(2,\ 976)} = 3.26$, $p \leq .04$; however, PSL differed significantly between species, Welch $F_{(2,\ 976)} = 4.88$, $p \leq .008$. Review of the post hoc analysis indicated that the African lion PSL was significantly larger than that of both the African leopard ($p \leq 6.83 \times 10^{-14}$) and the cheetah ($p \leq 1.42 \times 10^{-13}$). For M1 gigantopyramidal neurons, there were significant species differences in GSL, $F_{(2,\ 932)} = 18.83$, $p \leq 9.61 \times 10^{-9}$, GSA, $F_{(2,\ 932)} = 15.51$, $p \leq 2.3 \times 10^{-7}$, and GSV, $F_{(2,\ 932)} = 18.24$, $p \leq 1.69 \times 10^{-8}$. Examination of the post hoc analysis revealed that GSL, GSA, and GSV in the African leopard were significantly lower than those in the African lions, GSL: $p \leq 1.33 \times 10^{-10}$; GSA: $p \leq 9.60 \times 10^{-13}$; GSV: $p \leq 6.83 \times 10^{-14}$,

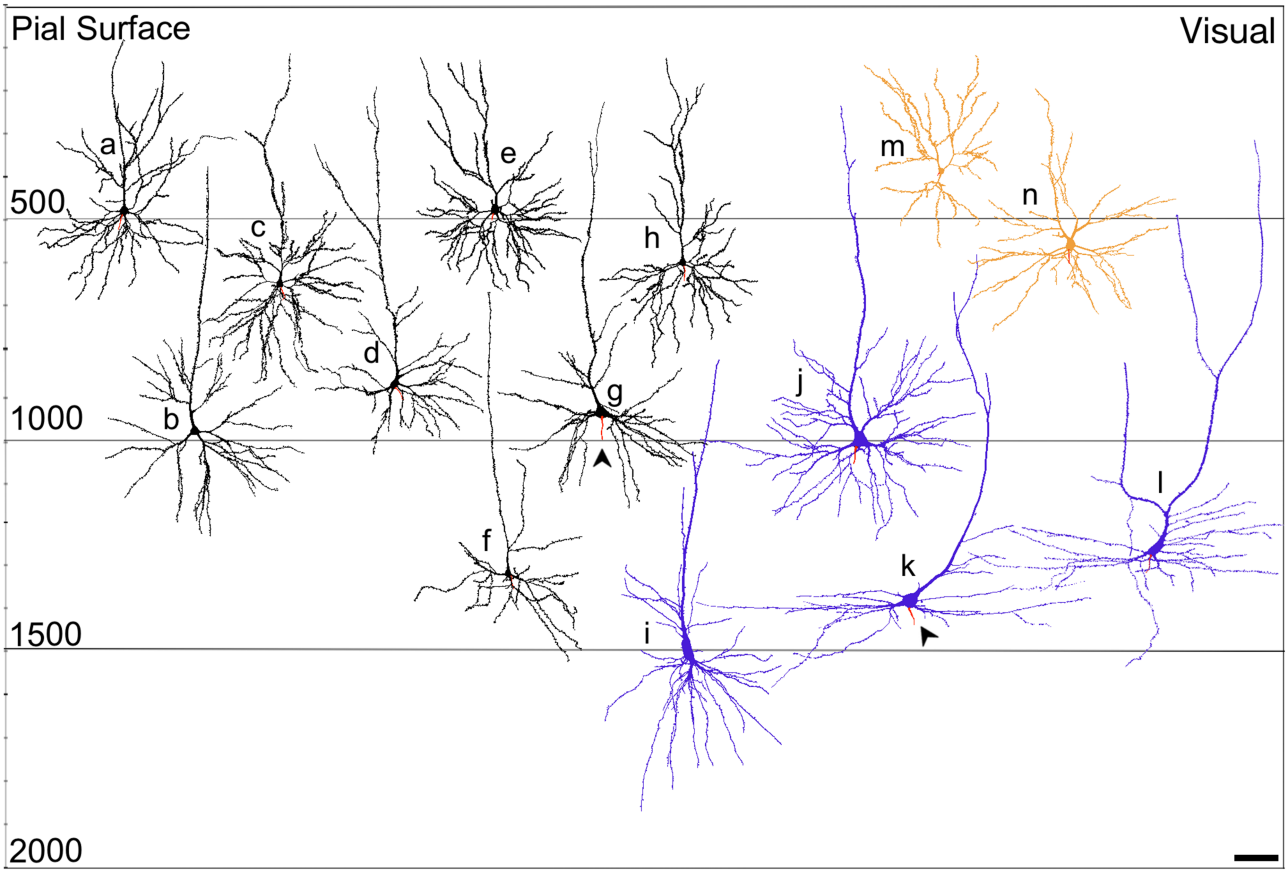

FIGURE 10 Neurolucida tracings of spiny neurons in the primary visual cortex of the cheetah indicating relative soma depth from the pial surface (in μm): Typical pyramidal neurons (a–h), solitary gigantopyramidal neurons of Meynert (i–l), and extraverted pyramidal neurons (m, n). Arrowheads indicate axons. Scale bar = 100 μm [Color figure can be viewed at wileyonlinelibrary.com]

and cheetah, GSL: $p \leq 4.87 \times 10^{-8}$; GSA: $p \leq 2.63 \times 10^{-11}$; GSV: $p \leq 1.42 \times 10^{-13}$.

For V1, significant species differences were identified for PSL, Welch $F_{(2,446)} = 24.07$, $p \leq 3.93 \times 10^{-10}$, PSA, Welch $F_{(2,\ 446)} = 25.56$, $p \leq 5.71 \times 10^{-10}$, and PSV, Welch $F_{(2,\ 446)} = 21.17$, $p \leq 3.93 \times 10^{-9}$. Review of the post hoc analysis indicated that the African lion PSV was significantly larger than that of the cheetah ($p \leq 1.47 \times 10^{-7}$) and African leopard ($p \leq .03$), whereas the cheetah had a significantly smaller PSL ($p \leq 4.17 \times 10^{-9}$). Significant differences were also observed in the size parameters of the gigantopyramidal neurons of Meynert, with species differences identified for GSL, Welch $F_{(2,\ 251)} = 14.5$, $p \leq 3.41 \times 10^{-6}$, GSA, Welch $F_{(2,251)} = 29.63$, $p \leq 8.14 \times 10^{-11}$, and GSV, Welch $F_{(2,\ 251)} = 40.44$, $p \leq 7.35 \times 10^{-14}$. Examination of the post hoc analysis showed that GSL was significantly lower in the African leopard in comparison to the cheetah ($p \leq .001$) and African lion ($p \leq .006$). GSA in the African leopard was also significantly lower than in the cheetah ($p \leq 4.70 \times 10^{-6}$) and the African lion ($p \leq 3.04 \times 10^{-5}$). Finally, GSV in the African leopard was significantly lower than in the cheetah ($p \leq 3.84 \times 10^{-8}$) and the African lion ($p \leq 4.27 \times 10^{-8}$).

For the prefrontal cortex, significant species differences were identified for PSL, Welch $F_{(2,883)} = 31.38$, $p \leq 9.58 \times 10^{-13}$, PSA, Welch $F_{(2,\ 883)} = 39.94$, $p \leq 7.53 \times 10^{-16}$, and PSV, Welch $F_{(2,\ 883)} = 47.17$, $p \leq 9.77 \times 10^{-19}$, with post hoc testing indicating that these parameters were markedly larger in the African lion.

Review of the OLS regression results indicated a hypometric (i.e., slope less than one; PSV is increasing at a slower rate that brain mass) relationship between brain mass and PSV (slope = 0.13, 95% CI slope = 0.04–0.21, Intercept = 3.29; $p \leq .01$; Figure 16a). Brain mass explained 46% of the variance in PSV across Carnivora. Review of the OLS regression results of GSV against brain mass revealed a hypometric scaling relationship similar to that seen for the pyramidal neurons. However, the data points were scattered about the line and the bivariate relationship was predictively weak. Brain mass only accounted for 19% of the total variance in GSV, and the relationship was not significant (slope = 0.26, 95% CI slope = –0.10–0.79, Intercept = 3.78; $p \leq .13$; Figure 16b). No phylogenetic signal was observed for the regression of PSV plotted against Brain mass (i.e., $\lambda < 0$). However, a phylogenetic signal was observed for GSV plotted against Brain mass although phylogenetic correction had no marked effect on the scaling parameters (i.e., $\lambda = 0.93$, slope = 0.29, Intercept = 3.61; $p \leq .08$, $R^2 = 0.25$). It would be inappropriate to overinterpret the GSV data given that the regression slope was not significant. Nevertheless, note that the felids were generally clustered together and that the African lion, cheetah, and African leopard

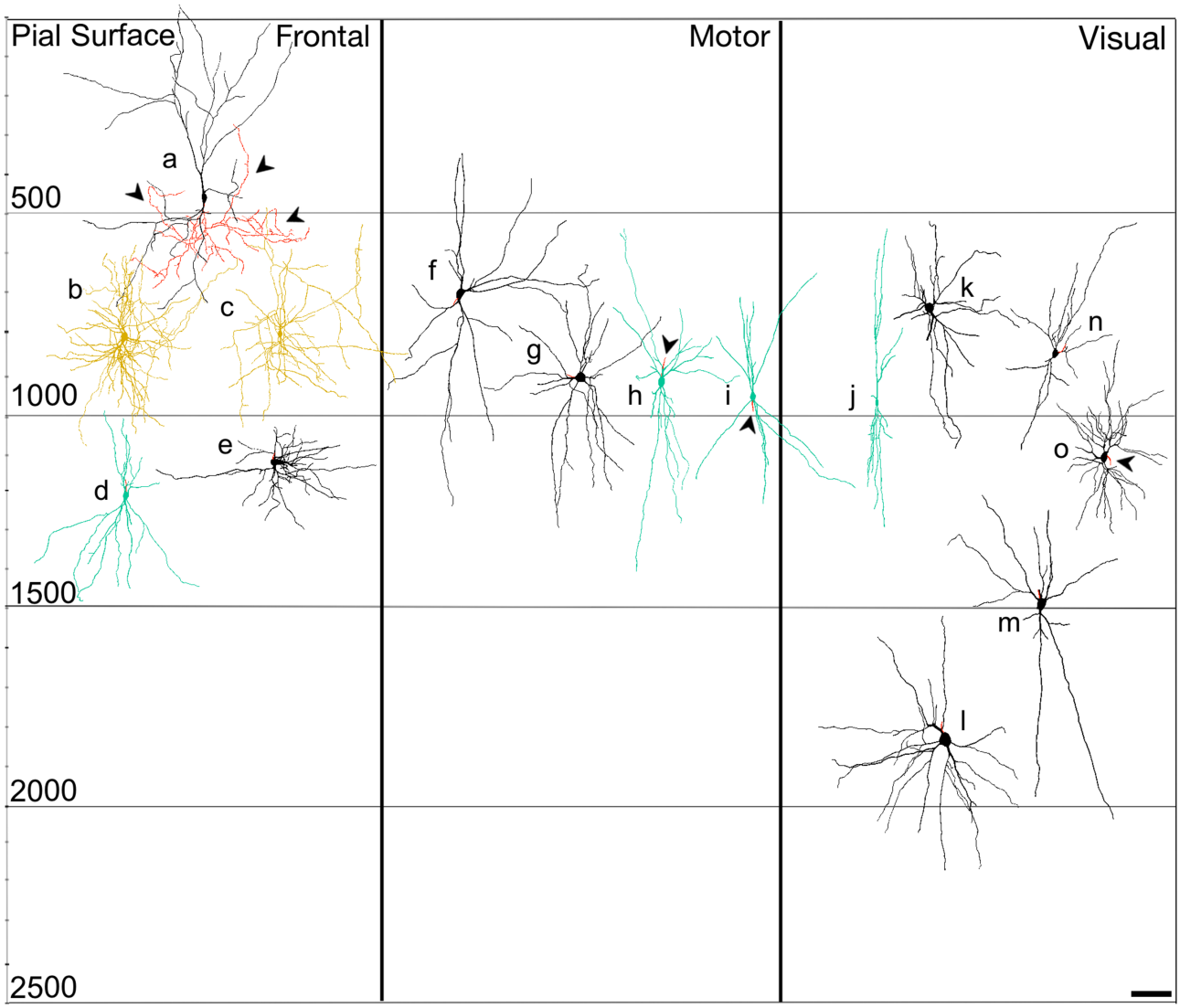

FIGURE 11 Neurolucida tracings of aspiny interneurons in the prefrontal (a–e), primary motor (f–i), and primary visual (j–o) cortices of the cheetah indicating relative soma depth from the pial surface (in μm): Aspiny multipolar (a, e–g, k–o), neurogliaform (b, c), and aspiny bipolar (d, h–j) neurons. Scale bar = 100 μm [Color figure can be viewed at wileyonlinelibrary.com]

data points lie above the regression line (outside of the confidence interval), suggesting a relatively larger GSV than expected based on brain size alone (Figure 16b).

4 | DISCUSSION

4.1 | Overview

In general, the neuronal types observed in the African lion, African leopard, and cheetah appeared consistent with those in other nondomestic felids (Jacobs et al., 2018; Johnson et al., 2016). Although there were morphological similarities across neurons in the three species studied, those traced in the cheetah tended to have more complex dendrites than those in the African leopard, despite their similar brain masses. The cheetah neurons also appeared to exhibit clear regional dendritic differences, unlike the African leopard neurons or those observed in other large felids (Johnson et al., 2016). Examination of the stereological results underscored the presence of very large gigantopyramidal neurons in layer V of M1. Comparative implications in the present study are limited by several methodological issues outlined below.

4.2 | Methodological considerations

The general limitations of Golgi research are well-documented (Braak & Braak, 1985; Horner & Arbuthnott, 1991; Williams, Ferrante, & Caviness, 1978). As noted by Arnold Scheibel, it is "one of the only histological techniques with personality" (Scheibel & Scheibel, 1978, p. 90), meaning that one cannot predict the outcome of any given stain (Friedland, Los, & Ryugo, 2006). Moreover, the stain is sensitive to postmortem delay variations, as well as types and length of fixation (Butti et al., 2015; Jacobs

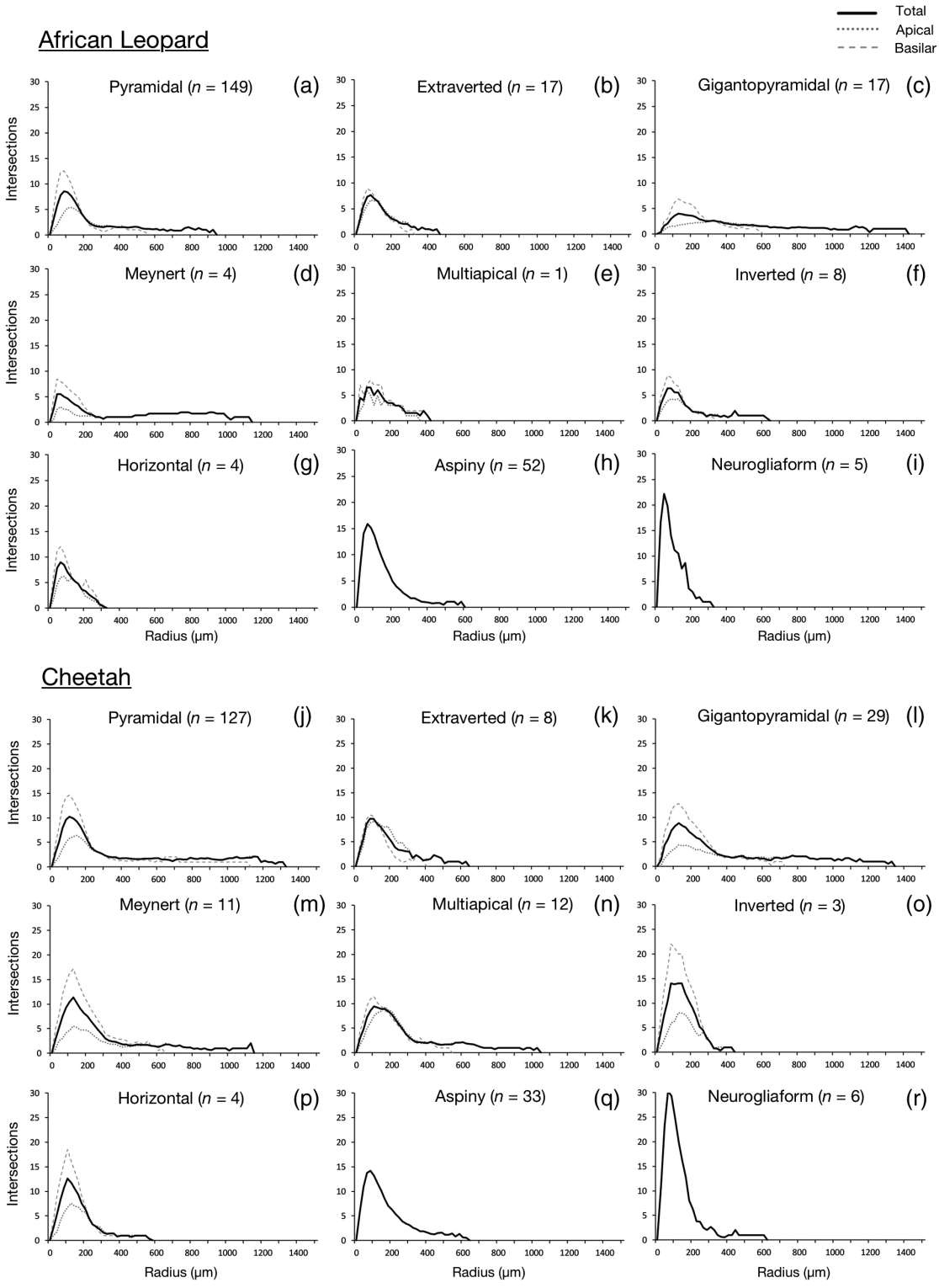

FIGURE 12 Sholl analyses indicating the relative branching complexity of apical, basilar, and total dendritic measurements for the African leopard and cheetah neurons. The Sholl analysis quantified dendritic branching at 20-μm intervals radiating away from the soma. Sholl analyses were not used for African lion neurons because of relatively incomplete impregnation of the Golgi stain. Branch intersections generally peaked near the soma (<200 μm) across all neuronal types. Peak intersections and overall dendritic complexity appeared to be greater for all neuron types (except aspiny interneurons) in the cheetah than in the African leopard

et al., 2001; Jacobs et al., 2018; Johnson et al., 2016). Especially relevant for quantitative investigations is the fact that dendritic measures (especially in larger neurons) are usually attenuated because of the 120 μm section thickness (Anderson et al., 2009; Jacobs et al., 2014; Jacobs & Scheibel, 1993), and that spine measures in light microscopy are always underestimations (Jacobs et al., 2001; Jacobs et al., 2011). Accepting

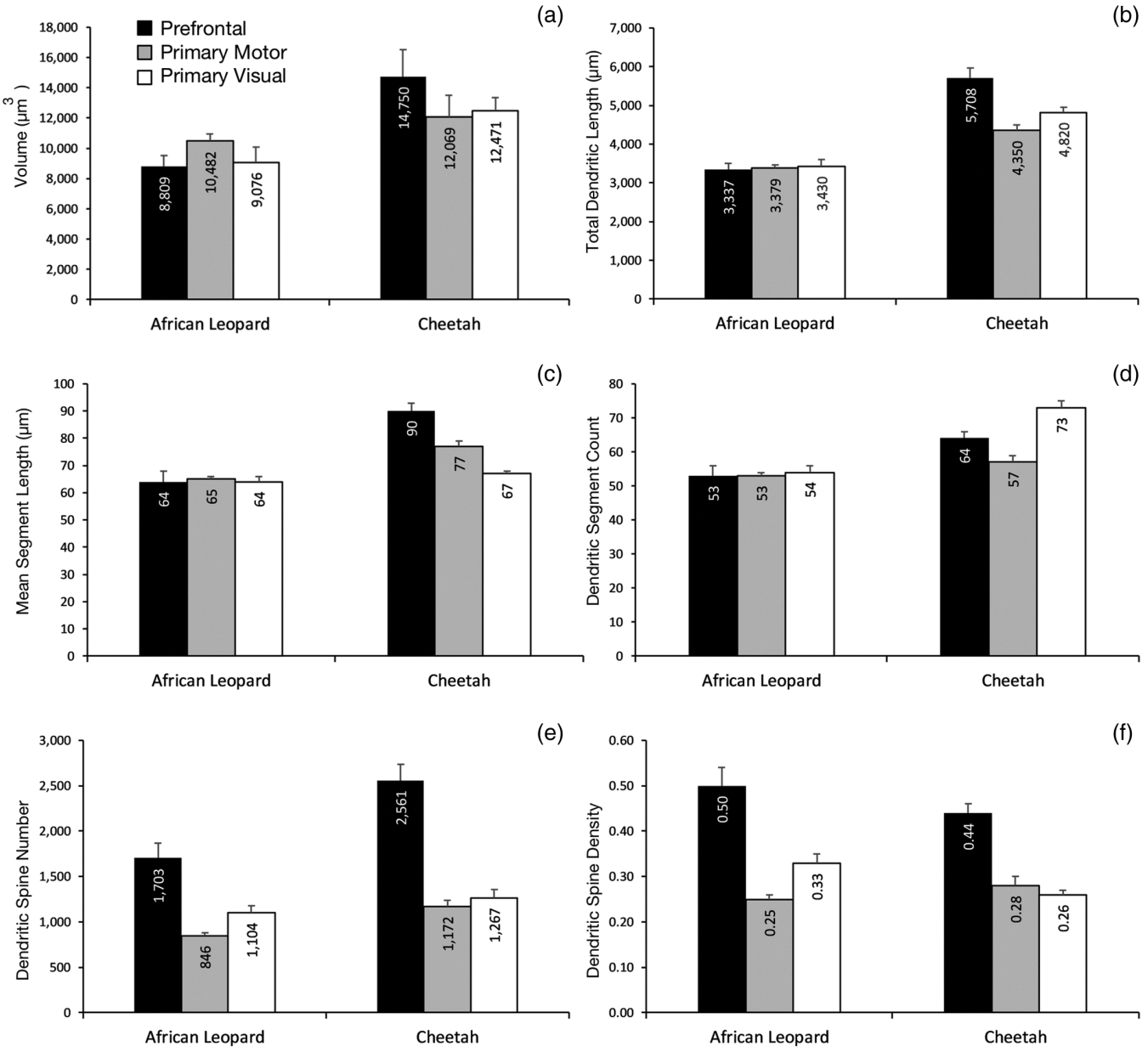

FIGURE 13 Bar graphs depicting the mean values of all dependent measures (a, Vol; b, TDL; c, MSL; d, DSC; e, DSN; f, DSD) of typical pyramidal neurons across the prefrontal, primary motor, and primary visual cortices in the African leopard and cheetah. Typical pyramidal neurons were generally more dendritically complex in the cheetah than in the African leopard. Pyramidal neurons also exhibited higher dendritic measures overall in the frontal lobe than in the other cortical regions in the cheetah, whereas the African leopard neurons did not show consistent regional differences. Error bars represent *SEM*

these general characteristics of the Golgi technique, there are also several important issues specific to the present article.

First, the relatively poor Golgi stain quality in the lion tissue, perhaps because of a prolonged storage time, prohibited quantitative comparisons with the other felids. The necessity for this exclusion becomes clear when one compares the average TDL values of M1 pyramidal neurons (5,399 ± 312 μm; soma depth = 1,087 ± 78 μm) in the lion from Jacobs et al. (2018) to the average TDL values obtained in the current study (4,135 ± 187 μm; soma depth = 984 ± 87 μm). The discrepancy is even greater for M1 gigantopyramidal neurons, with TDL at 8,600 ± 814 μm in Jacobs et al. (2018) but only 1,872 ± 215 μm in the current study. Nevertheless, the broader sampling of neuron types (Figure 5) in the present study provides a valuable supplement to the pyramidal and gigantopyramidal tracings in Jacobs et al. (2018).

Second, neuromorphological explorations of rare species (all of the felids in the present study have a conservation status of "vulnerable") are, with only one or two specimens per species, essentially case studies. As such, although qualitative classifications of neuronal morphology for a given species may be representative of the species, more detailed quantitative comparisons remain problematic. Thus, it is possible that the available tissue specimens in such "case studies" may not be quantitatively representative of the species. Specifically, in the present study,

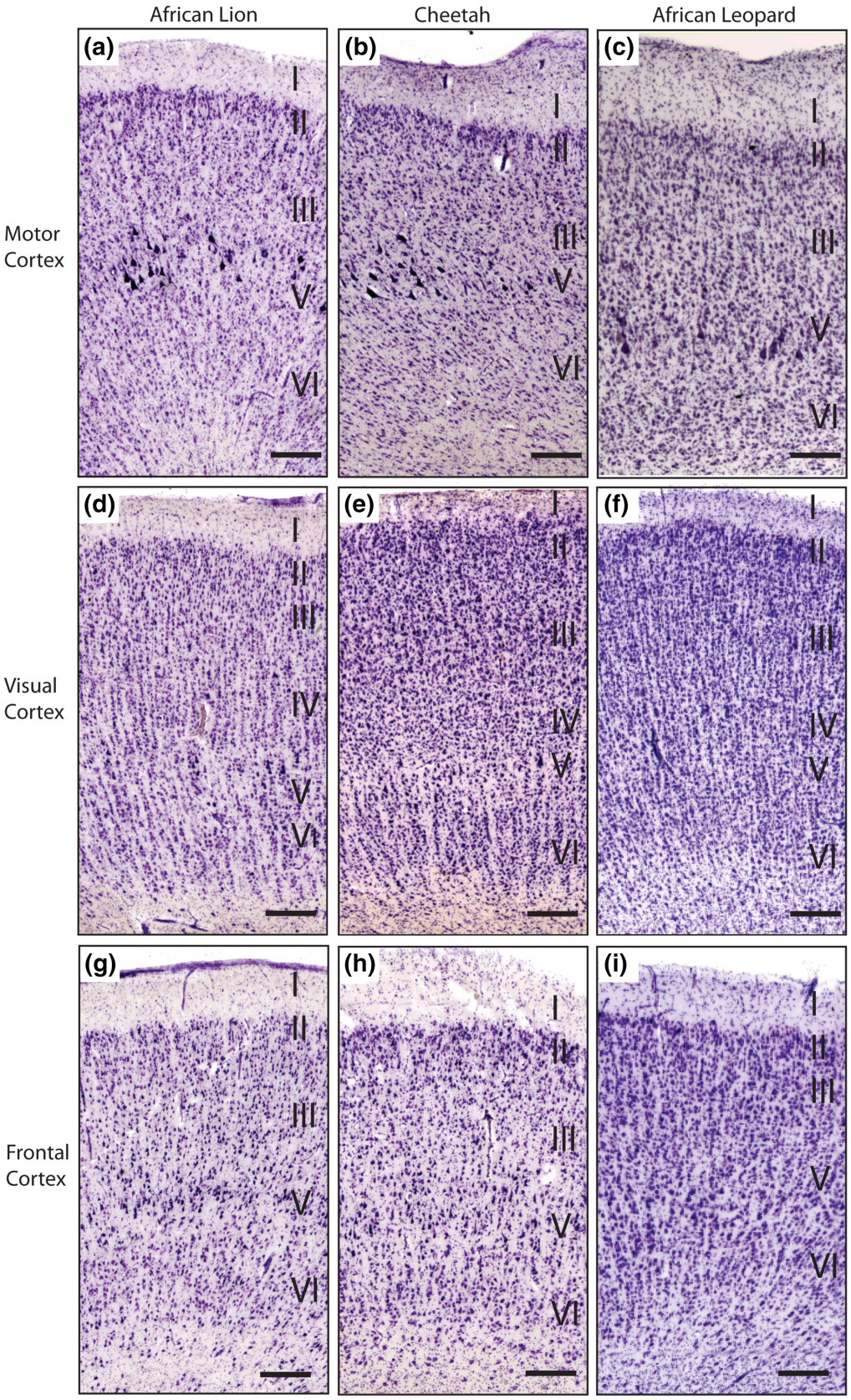

FIGURE 14 Photomicrographs of Nissl-stained tissue in the primary motor, visual, and frontal cortices of the African lion, cheetah, and African leopard. Cortical layers are labeled with Roman numerals. Note the large gigantopyramidal neuron somata present in layer V of the primary motor cortices for all three species. Scale bar = 100 μm [Color figure can be viewed at wileyonlinelibrary.com]

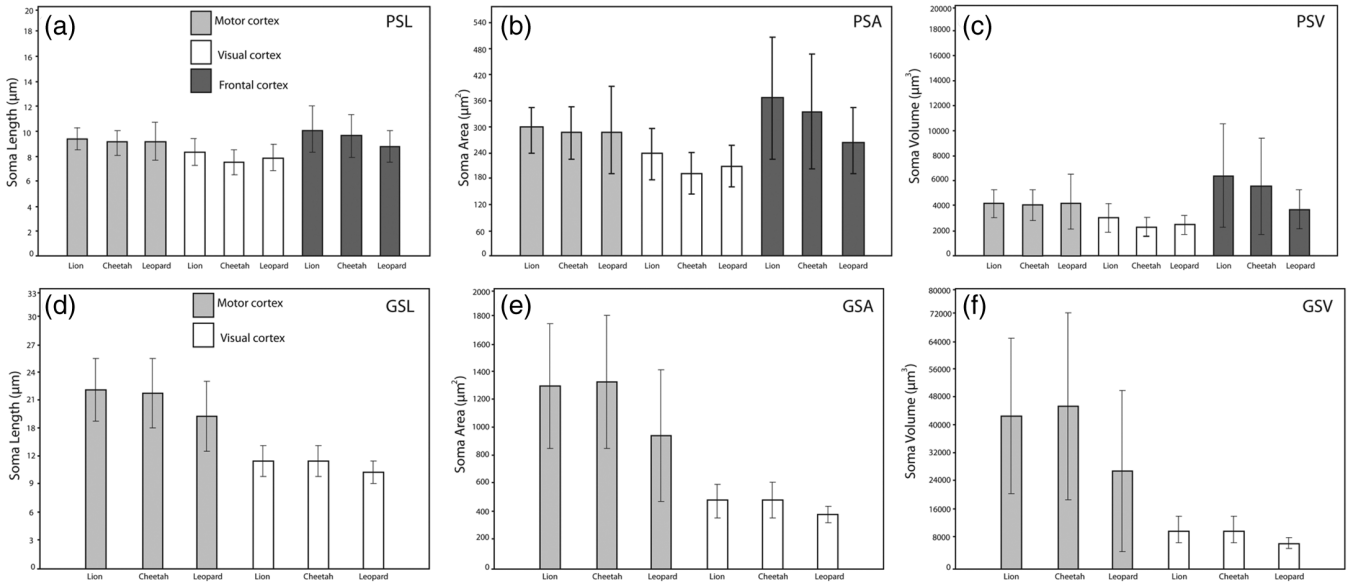

FIGURE 15 Bar graphs showing the mean and *SD* of neuronal soma parameters in sampled cortical areas (primary motor, primary visual, and prefrontal) of the African lion, African leopard, and cheetah. Graphs a–c represent pyramidal neurons; graphs d–f represent gigantopyramidal neurons. Measures are soma length (PSL in a; GSL in d), soma area (PSA in b; GSA in e), and soma volume (PSV in c; GSV in f). In all three species, PSV (c) was generally larger in the frontal cortex followed by the primary motor and then visual cortices. Gigantopyramidal neurons tended to be largest in the lion and the cheetah, and were substantially larger in primary motor cortex than in the visual cortex for all three species

individual variation in behavior, breeding status, sex, and captivity of cheetahs may be reflected in the quantitative measures of the captive female cheetah specimen (Quirke, O'Riordan, & Davenport, 2013; Wielebnowski, 1999). Negative effects of aging on dendritic branching and spine density (Jacobs, Driscoll, & Schall, 1997; Jacobs & Scheibel, 1993) may have affected dendritic measurements in the 22-year-old African leopard specimen, which is much older than the average lifespan of wild African leopards (~7 to 12 years; Guggisberg, 1975). Despite these limitations, the availability of well-preserved tissue from such rarely studied species presented a unique opportunity to contribute to a comparative neuromorphological database for nondomestic felids.

In the sections that follow, we first discuss the general morphological findings for spiny (4.3) and aspiny (4.4) neurons. We then examine the quantitative somatodendritic findings for typical pyramidal neurons (4.5.1) and M1 gigantopyramidal neurons (4.5.2) vis-à-vis those of large felids documented in Johnson et al. (2016) and Jacobs et al. (2018). Finally, we discuss the stereological results (4.6) of large felids in comparison with those of other carnivores.

4.3 | Spiny neurons

4.3.1 | Typical pyramidal neurons

The average number of primary basilar dendrites per pyramidal neuron across the three species (~4.29) was similar to that in the Siberian tiger and clouded leopard (4.09; Johnson et al., 2016), but all five felids still had fewer than what has been reported by others using injection techniques in the domestic cat (Lucifer yellow: ~6 to 9, Hübener et al., 1990; Horseradish peroxidase: ~5.15, Ghosh, Fyffe, & Porter, 1988). As has been noted previously (Johnson et al., 2016), such differences in the basilar array may indicate a lower cell-packing density in larger felids than in the domestic cat (Jardim-Messeder et al., 2017). Similar to the Siberian tiger and clouded leopard (Johnson et al., 2016), and the domestic cat (Ghosh et al., 1988; Hübener et al., 1990; Yamamoto, Samejima, & Oka, 1987), most apical dendrites in the present study singularly ascended vertically from the soma, with some bifurcating narrowly in a V-shaped pattern a slight distance from the soma. Thus, the present findings suggest that felid neocortical architecture generally illustrates a mostly vertically-oriented type of columnar information processing similar to that documented in rodents and primates (DeFelipe, Hendry, Hashikawa, Molinari, & Jones, 1990; Innocenti & Vercelli, 2010; Mountcastle, 1997; Valverde & Facal-Valverde, 1986), with mini-columns similar to those documented in the giraffe, platypus, and human (DeFelipe, Alonso-Nanclares, & Arellano, 2002). It has been suggested that mini-columns, or infrabarrels in rodent somatosensory cortex, may be involved in corticothalamic integration (Crandall, Patrick, Cruikshank, & Connors, 2017). This vertical arrangement appears to differ from the more lateral projecting pyramidal neurons in cetartiodactyls (cetaceans: Butti et al., 2015; giraffe: Jacobs, Harland, et al., 2015; pygmy hippopotamus: Butti et al., 2014) and especially the African elephant (Jacobs et al., 2011). A more detailed comparative discussion of these neurons is provided below in Section 4.5.1.

4.3.2 | Extraverted pyramidal neurons

Extraverted neurons were prevalent in the superficial cortical layers in all three species. As indicated by review of the Sholl analyses, these were the only neuron type with apical branching that was equal to, or more

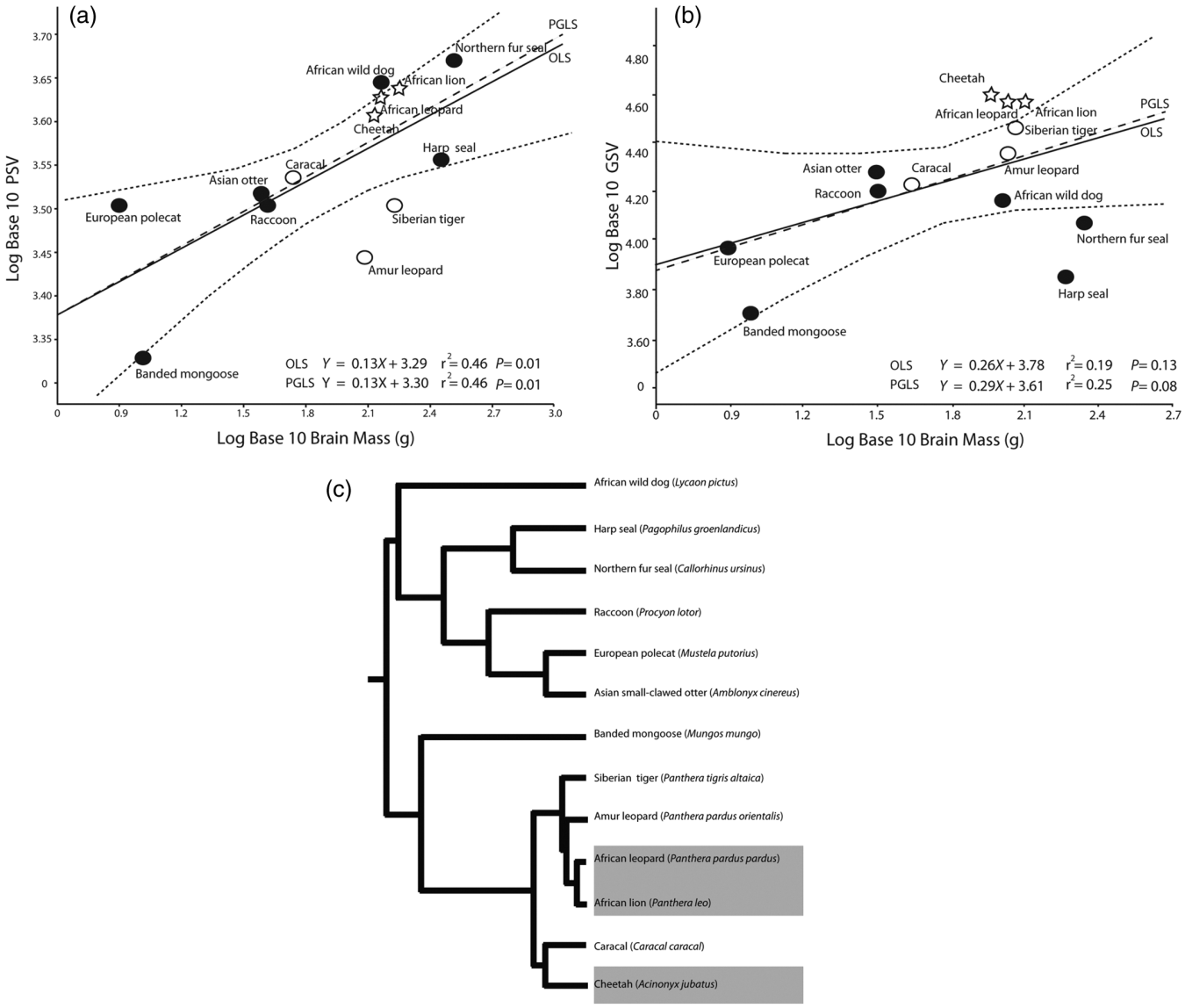

FIGURE 16 Scatterplots showing the bivariate linear regression analysis (Ordinary Least Squares, OLS, and Phylogenetic Generalized Least Squares, PGLS) of brain mass (grams) plotted against the parameters PSV (a, Pyramidal neuron Soma Volume) and GSV (b, M1 Gigantopyramidal neuron Soma Volume). The data were obtained from the current study (depicted by the star symbols), as well as from Jacobs et al. (2018), depicted as open circles for other feliforms and filled circles for non-felid carnivores. All data were log transformed (base 10) prior to being included in the analysis. In general, there was a hypometric relationship between brain mass and both PSV and GSV, with the felids generally clustered together for GSV. Figure 16c shows the phylogeny for the implementation of PGLS, with shaded regions highlighting the three species in the current study (see text for details)

extensive than, basilar branching. Extraverted neurons have previously been observed in certain species that have been considered "primitive" (Marín-Padilla, 1992), such as the hedgehog, as well as in the bat, opossum, rat, rhesus monkey, and domestic cat (Ferrer, 1987; Ferrer et al., 1986a, 1986b; Sanides & Sanides, 1972; Valverde & Facal-Valverde, 1986). However, recent research in larger mammals (e.g., the pygmy hippopotamus and cetaceans, Butti et al., 2014, 2015; Ferrer & Perera, 1988; Hof & Van der Gucht, 2007), suggests that this type of neuron may represent a different processing adaptation for mammalian species with agranular cortices. The presence of extraverted neurons in large feliforms (Johnson et al., 2016), including those in the present study, confirms that they are not exclusive to smaller "primitive" species, nor to species with agranular cortices. Quantitatively, extraverted neurons in the cheetah were generally larger than in the African leopard, but both were smaller than those in the Siberian tiger and larger than those in the clouded leopard in terms of Vol, TDL, and MSL (Johnson et al., 2016). However, the soma size of these neurons in the cheetah and African leopard were smaller than those observed in both the Siberian tiger (32% smaller in cheetah and 37% smaller in African leopard) and clouded leopard (7% smaller in cheetah and 15% smaller in African leopard; Johnson et al., 2016). Furthermore, extraverted neurons of the African leopard displayed the smallest DSC (28% lower than cheetah, 38% lower than tiger, 14% lower than clouded leopard; Johnson et al., 2016) and the fewest dendritic intersections in Sholl analyses of the felids, suggesting that these neurons in the African leopard had less branching than those in the other felids.

4.3.3 | Gigantopyramidal neurons in M1

M1 gigantopyramidal neurons were the largest neuronal type traced across the three felid species, resembling human Betz cells in layer Vb of M1 (Betz, 1874; Braak & Braak, 1976; Meyer, 1987; Rivara, Sherwood, Bouras, & Hof, 2003). Betz (1874) originally observed M1 gigantopyramidal cells in canids and primates; subsequently, Brodmann (1909) observed that these neurons were especially large in carnivores such as the brown bear, wolf, tiger, and lion. In a recent 19-species comparative study, Jacobs et al. (2018) confirmed Brodmann's (1909) findings that gigantopyramidal motoneurons in feliforms (e.g., caracal and clouded leopard), particularly those within the *Panthera* genus (e.g., African lion and Siberian tiger), were disproportionally larger than those in noncarnivore species. The current results are consistent with those of Jacobs et al. (2018) insofar as large gigantopyramidal neurons were documented in M1 of both the African leopard (a *Panthera* species) and the cheetah (a non-*Panthera* species). These neurons were generally larger (as measured by dendritic Vol) in the African leopard than in the cheetah. Although the dendritic data from the lion in the current study were not used for quantitative comparisons, the size of M1 gigantopyramidal somata was still relatively large. A more detailed comparative discussion of these neurons is provided below in Section 4.5.2.

4.3.4 | Solitary gigantopyramidal neurons of Meynert

Solitary gigantopyramidal neurons of Meynert were identified in V1 of all three felid species. Previously classified as "magnopyramidal" neurons in the visual cortices of the domestic cat (Gabbott, Martin, & Whitteridge, 1987; Winer & Prieto, 2001) and other mammals (clouded leopard: Johnson et al., 2016; dolphin and minke whale: Butti et al., 2015; giraffe: Jacobs, Harland, et al., 2015), these neurons were morphologically similar to solitary cells of Meynert found in the primate visual cortex (Chan-Palay, Palay, & Billings-Gagliardi, 1974; Hof, Nimchinsky, Young, & Morrison, 2000; Meynert, 1867). Magnopyramidal neurons in V1 of the clouded leopard (Johnson et al., 2016) were 10% larger in soma cross-sectional area than solitary gigantopyramidal cells of Meynert in the African leopard, which were generally larger than in the cheetah. However, these neuron types in the cheetah had more complex dendrites than those in the African and clouded leopard (Johnson et al., 2016). The general incompleteness and small number of neuronal tracings impedes definitive conclusions. Nevertheless, these enlarged V1 pyramidal neurons may represent a predatory-related specialization of sensorimotor cortical areas for lions, leopards, and cheetahs (Sherwood et al., 2003), allowing them to integrate broad-range visual information to detect prey over wide, open terrains (Kleiman & Eisenberg, 1973; Martins & Harris, 2013).

4.3.5 | Atypical pyramidal neurons

The small number of quantified atypical pyramidal neuron variants (e.g., multiapical, inverted, and horizontal pyramidal neurons) limits the extent of comparative inferences. For instance, only one multiapical pyramidal neuron was traced in the African leopard, three in the lions, and 12 in the cheetah. However, multiapical neurons have also been documented in the clouded leopard (Johnson et al., 2016) and in several non-feline species (African elephant: Jacobs et al., 2011; bat: Ferrer, 1987; bottlenose dolphin and humpback whale: Butti et al., 2015; rat and hedgehog: Ferrer et al., 1986a). Inverted pyramidal neurons, historically believed to be the result of faulty neuronal migration (Van der Loos, 1965), appear to contribute to long-range corticostriatal, corticoclaustral, and corticocortical communication (Mendizabal-Zubiaga, Reblet, & Bueno-Lopez, 2007). Inverted pyramidal neurons were present in all three felids examined. They have been documented in small numbers across multiple species (African elephant: Jacobs et al., 2011; chimpanzee: Qi, Jain, Preuss, & Kaas, 1999; domestic cat: Ferrer et al., 1986b; Matsubara et al., 1996; giraffe: Jacobs, Harland, et al., 2015; minke whale: Butti et al., 2015). Inverted pyramidal neurons in the cheetah were consistently more dendritically complex than those in the African leopard, and were also larger than those in the Siberian tiger and clouded leopard (Johnson et al., 2016). Finally, horizontal pyramidal neurons were documented in both the cheetah and the African leopard. These neurons appear to facilitate the lateral integration of cortical information (Van Brederode, Foehring, & Spain, 2000) and have been traced in various cortical layers in the Siberian tiger and clouded leopard (Johnson et al., 2016) and many other mammalian species (African elephant: Jacobs et al., 2011; cetaceans: Butti et al., 2015; dog and cat: Ferrer et al., 1986b; rodents: Ferrer et al., 1986a; giraffe: Jacobs, Harland, et al., 2015; primates: de Lima, Voigt, & Morrison, 1990; Meyer, 1987). Horizontal neurons in the African leopard were somewhat larger than in both the Siberian tiger and the clouded leopard (Johnson et al., 2016), but these neurons were all smaller than in the cheetah. In terms of TDL, horizontal neurons in the cheetah were comparable to those reported in the giraffe (Jacobs, Harland, et al., 2015) as well as the minke and humpback whales (Butti et al., 2015), animals with much larger brain/body sizes.

4.4 | Aspiny interneurons

Aspiny interneurons are a diverse group of inhibitory cortical neurons (Mihaljević, Benavides-Piccione, Bielza, Larrañaga, & DeFelipe, 2019; Petilla Interneuron Nomenclature Group, 2008; Sherwood et al., 2009). In the present study, they exhibited comparable somatodendritic values in the cheetah and African leopard, both falling in size between the larger interneurons in the Siberian tiger and the smaller interneurons in the clouded leopard (Johnson et al., 2016). Sholl analyses indicated that the dendritic branching of aspiny neurons peaked ~600 μm from the soma in both the African leopard and cheetah, extending further than in the clouded leopard (~500 μm; Johnson et al., 2016) but less than in the Siberian tiger (~800 μm; Johnson et al., 2016). Finally, neurogliaform cells, which may establish electrical coupling and synchronous inhibition across neuronal circuits (Simon, Oláh, Molnár, Szabadics, & Tamás, 2005; Zsiros & Maccaferri, 2005), possessed the most complex dendritic plexus of all neuronal types. Although neurogliaform cells in the cheetah

TABLE 3 Demographic information for felids in quantitative comparisons arranged by decreasing brain mass

	Siberian Tiger[a]	African Lion[b]	African Leopard	Cheetah	Clouded Leopard[a]	Caracal[b]
	Panthera tigris altaica	*Panthera leo*	*Panthera pardus pardus*	*Acinonyx jubatus*	*Neofelis nebulosa*	*Caracal caracal*
Age (years)/sex	12/female	Adult/female	22/male	Adult/female	20 and 28/female (n = 2)	Adult/male
Fixation method	In situ perfusion	In situ perfusion	In situ perfusion	In situ perfusion	Immersion	In situ perfusion
Regions sampled[c]	LH frontal, motor, and visual	LH motor	LH frontal, motor, and visual	LH frontal, motor, and visual	RH frontal, motor, and visual	LH motor
Storage time (months)[d]	6	14	6	18	5 and 36	21
Brain mass (g)	258	213	143	140	82 and 73	58
Number of neurons traced						
Pyramidal	34	20	149	127	71	19
[e]*Gigantopyramidal*	3	10	17	29	3	10
Soma depth (μm)[f]						
Pyramidal	939 ± 53	1,087 ± 78	896 ± 30	880 ± 33	898 ± 43	953 ± 81
[a]*Gigantopyramidal*	1,573 ± 64	1,558 ± 40	1,467 ± 40	1,637 ± 29	1,414 ± 79	1,199 ± 47
Soma size (μm^2)[f]						
Pyramidal	454 ± 49	441 ± 50	274 ± 9	388 ± 24	276 ± 10	407 ± 27
[a]*Gigantopyramidal*	4,028 ± 113	2,824 ± 284	2,548 ± 158	2,210 ± 163	3,882 ± 444	2,000 ± 182

Note: All specimens were obtained following euthanasia with an autolysis time < 30 min.
[a]From Johnson et al. (2016).
[b]From Jacobs et al. (2018).
[c]LH, left hemisphere; RH, right hemisphere.
[d]Time fixed tissue was stored in 0.1% sodium azide before sectioning.
[e]Gigantopyramidal neurons in primary motor cortex.
[f]Soma depth from the pial surface and soma size of neurons represented by mean ± *SEM*.

were generally smaller than in the African leopard, they were more dendritically complex, possessing a higher magnitude of dendritic intersections in the Sholl analyses than those in African leopard, Siberian tiger, and clouded leopard (Johnson et al., 2016). In fact, cheetah neurogliaform neurons approached the complexity of those observed in the giraffe, an animal with a much larger brain (Jacobs, Harland, et al., 2015).

4.5 | Quantitative somatodendritic comparisons across felids

To provide a more comprehensive understanding of the neuromorphological variation across felid species, the present results for typical pyramidal and M1 gigantopyramidal neurons were compared with those from several other large felids quantified previously with similar Golgi staining and reconstruction methodologies (Siberian tiger and clouded leopard from Johnson et al., 2016; African lion and caracal from Jacobs et al., 2018). Despite the limitations of the Golgi stain, its consistent usage across many studies of different species allows for the most inclusive cross-species data collection strategy, facilitating comparisons of similar neuron types despite the relatively low number of specimens (e.g., African elephant and human, Jacobs et al., 2011; African elephant and select cetartiodactyls, Jacobs, Harland, et al., 2015). Summary demographic information for all species and the number of neurons traced are provided in Table 3.

4.5.1 | Typical pyramidal neurons

Representative tracings of pyramidal neurons are presented along with quantitative comparisons of dendritic (Vol, TDL, MSL, DSC) and spine measures (DSN, DSD) across the six felids in Figure 17. When comparing the *Panthera* species and the closely-related clouded leopard, dendritic measures generally increased with brain mass as expected (Siberian tiger > African lion > African leopard > clouded leopard), with the largest felid, the Siberian tiger, exhibiting the most complex pyramidal neurons (in Vol, TDL, MSL, and DSC). In contrast, the two distant non-*Panthera* felids, the cheetah and caracal, did not exhibit a clear relationship between dendritic extent and brain size. Typical pyramidal neurons in the cheetah exhibited more complex dendritic arborizations than in the African leopard, a felid with a similar brain and body size, for all parameters except DSD. As highlighted in Figure 13, the prefrontal cortex of the cheetah, a semisocial felid (Caro & Collins, 1987), exhibited more complex neurons than did M1 and V1. Conversely, the African leopard and previously observed nondomestic felids (Siberian tiger and clouded leopard: Johnson et al., 2016) exhibited no consistent regional pattern in dendritic extent. These results suggest that, although brain and body size may affect dendritic extent, there may be other effects such as age (Jacobs et al., 1997), environmental enrichment (Diamond, 1967; Jacobs, Schall, & Scheibel, 1993), and/or social complexity, which has been

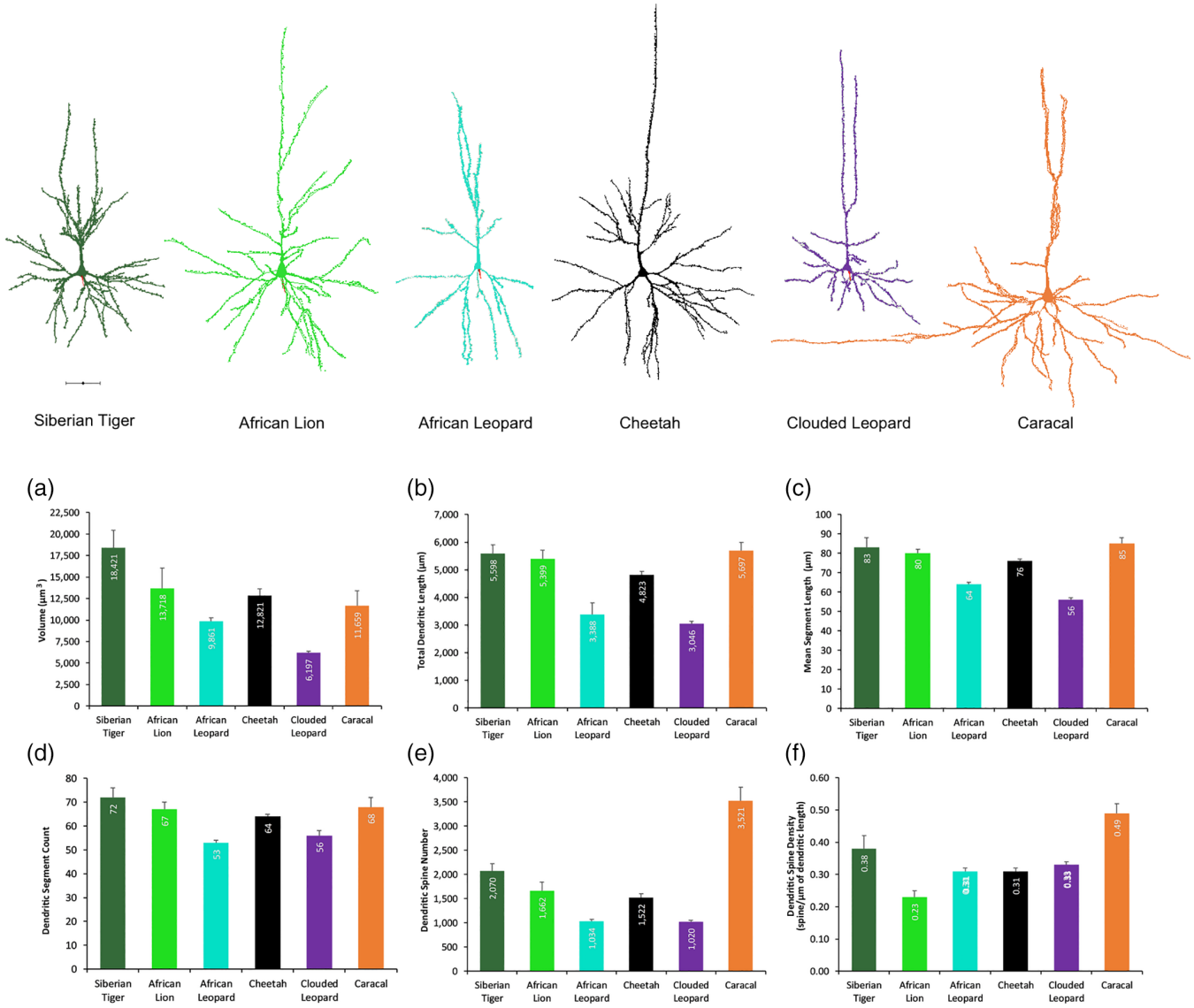

FIGURE 17 Qualitative and quantitative comparisons of typical pyramidal neurons across six felids arranged by decreasing brain mass (see Table 3): Siberian tiger, African lion, African leopard, cheetah, clouded leopard, and caracal. The displayed pyramidal neuron tracings were chosen for completeness of Golgi impregnation and represent the approximate average TDL for each species. Bar graphs present comparative data of typical pyramidal neurons for the mean values of six dependent measures (a, Vol; b, TDL; c, MSL; d, DSC; e, DSN; f, DSD). Note that dendritic measures of typical pyramidal neurons in cheetah and caracal do not positively correlate to brain mass as observed in the other species. Error bars represent *SEM*. Scale bar = 100 µm [Color figure can be viewed at wileyonlinelibrary.com]

previously associated with relative neocortical size (Dunbar, 1998; Dunbar & Bever, 1998).

Furthermore, the caracal possessed the smallest brain size, yet its pyramidal neurons were some of the largest and most dendritically complex among all felids. The caracal exhibited relatively high Vol and DSC. It also had the highest TDL and MSL, similar to those of the Siberian tiger, an animal with a brain size approximately five times larger than the caracal (Table 3). Spine measures in the caracal were also high, perhaps due to a high quality of Golgi stain. However, the other disproportionate dendritic measures are more difficult to explain. The phylogenetic background of caracal may contribute to this dendritic discrepancy insofar as the caracal is one of the earliest (~8.5 Ma, Johnson et al., 2006) migrators to Africa (Figure 16c) and remains relatively restricted to Africa (O'Brien & Johnson, 2005). The cheetah, the other felid that exhibited higher dendritic measures than expected, also experienced a migration that separated its species from its evolutionary relatives. The cheetah remains particularly genetically homogenous because of severe population bottlenecks (Dobrynin et al., 2015; O'Brien et al., 1985), causing its relationship to other felids to be even more distant and uncertain (Bininda-Emonds et al., 2001; Bininda-Emonds, Gittleman, & Purvis, 1999). Although larger sample sizes are clearly necessary for any definitive conclusion, these divergences from the *Panthera* genus suggest that the high neuronal complexity in caracal and cheetah may be a result of convergent evolution as they evolved independently from their *Panthera* relatives.

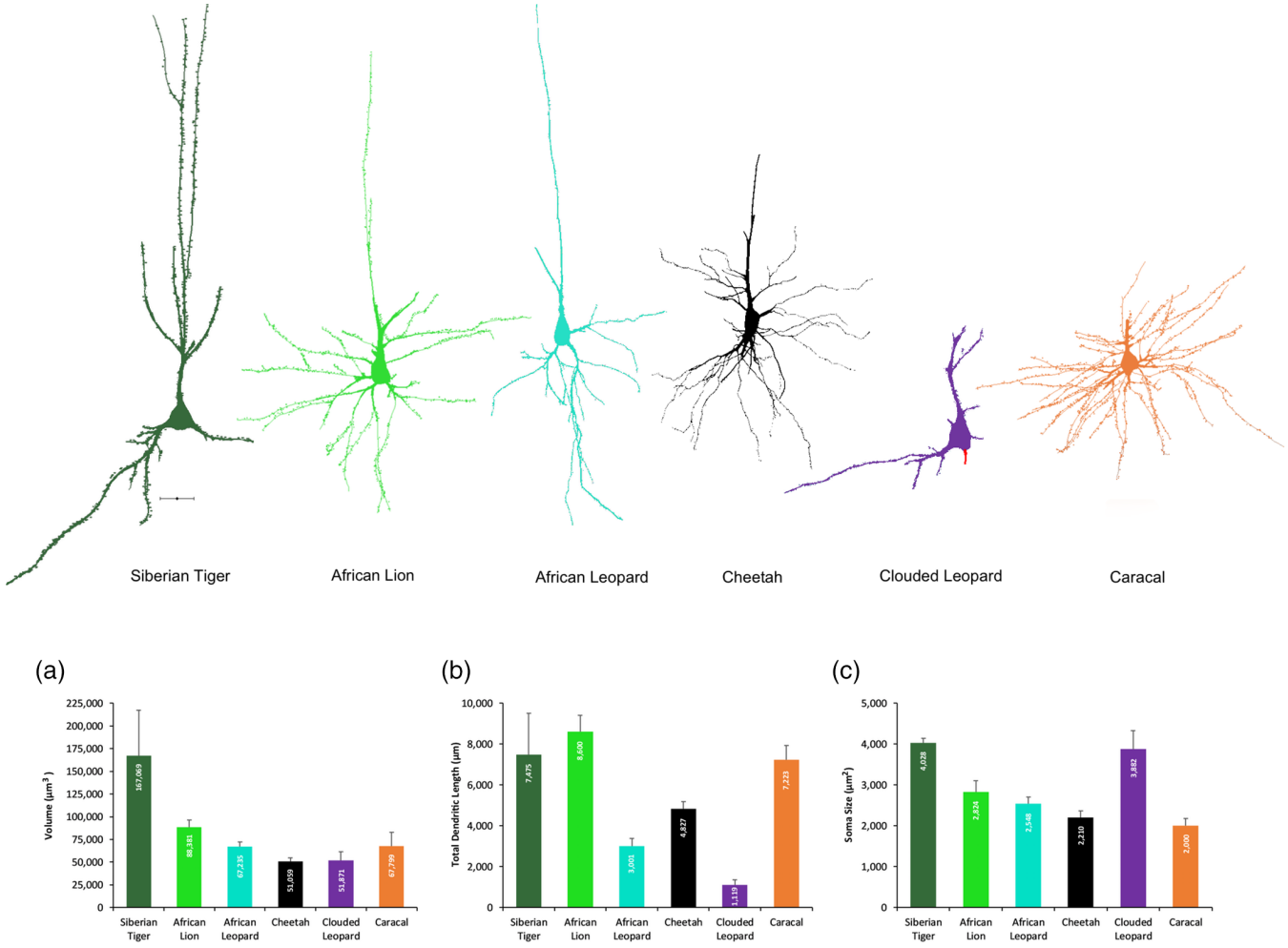

FIGURE 18 Qualitative and quantitative comparisons of primary motor gigantopyramidal neurons across six felids arranged by decreasing brain mass (see Table 3): Siberian tiger, African lion, African leopard, cheetah, clouded leopard, and caracal. The displayed gigantopyramidal neuron tracings represent the approximate average soma size for each species. Bar graphs present comparative data of these neurons for mean values of three dependent measures (a, Vol; b, TDL; c, soma size). Note the extreme truncation of primary motor gigantopyramidal neuron tracings, particularly in the clouded leopard. Error bars represent *SEM*. Scale bar = 100 µm [Color figure can be viewed at wileyonlinelibrary.com]

4.5.2 | M1 gigantopyramidal neurons

Representative tracings of M1 gigantopyramidal neurons for each felid, and quantitative comparisons for dendritic Vol, TDL, and soma size are provided in Figure 18. Because these tracings remain incomplete due to tissue sectioning, other dendritic measures were excluded from this comparison. As with pyramidal neurons, dendritic Vol in M1 gigantopyramidal neurons generally increased with brain size for *Panthera* feliforms and for the closely-related clouded leopard, but not for the cheetah and caracal. Notably, the cheetah M1 gigantopyramidal neurons exhibited the smallest dendritic Vol of the six species. TDL had no consistent pattern across the felids, with the African lion having the largest TDL and caracal and Siberian tiger having comparable measures. TDL of the clouded leopard M1 gigantopyramidal neurons was noticeably smaller than in the other species, perhaps because it was immersion-fixed whereas the other species were perfused (Table 3). However, despite having the lowest TDL of the felids, the clouded leopard possessed extremely large soma sizes, second only to the Siberian tiger. According to the morphological analysis, M1 gigantopyramidal somata from the cheetah and caracal were smaller than those from the other felids. These inconsistencies illustrate the variability in gigantopyramidal tracings across different species. However, the more comprehensive stereological analysis revealed that M1 gigantopyramidal neuron somata were exceptionally large for all three felids compared to those of M1 gigantopyramidal neurons in several non-feline species with similar or larger brains (Figure 16). The large size of M1 gigantopyramidal neurons in the non-*Panthera* cheetah, as well as the *Panthera* species, suggests that these cell types may have evolutionarily converged within the Felidae family.

The functional implications for gigantopyramidal neurons in M1 remain unclear. As suggested by Jacobs et al. (2018), the large size of these neuron types may contribute to the innervation of fast-fatiguing Type IIx muscle fibers specialized for stalking and pouncing behavior in felids, as has been documented in the African lion and caracal (Kohn, Burroughs, Hartman, & Noakes, 2011; Kohn & Noakes, 2013). Although the cheetah is not an ambush predator that attacks at close range like

other felids, it is the fastest extant terrestrial mammal and captures its prey through high-speed chases over several hundred meters (Eaton, 1970). The cheetah excels in speed and accelerates with power up to 120 W/kg, then absorbs the shock to decelerate almost as effectively and make sharp turns towards prey (Kaplan, 2013; Wilson et al., 2013). This impressive balance of speed, acceleration, and turning maneuverability is achieved by the cheetah's disproportionally large hindlegs (Hudson et al., 2010, 2011), which are composed of predominantly Type IIb muscle fibers, an MHC isoform capable of exerting the largest force of all Type IIx fibers (Goto et al., 2013; Hyatt, Roy, Rugg, & Talmadge, 2010). Because specialized Type IIx muscle fiber compositions require innervation by larger motoneurons (Goto et al., 2013; Kohn et al., 2011), M1 gigantopyramidal neurons may have increased in size to adapt to the specialized physical demands of the cheetah, as has been previously proposed for other felids (Jacobs et al., 2018).

4.6 | Stereological analysis

Although examination of the stereological results revealed similarities across the prefrontal, motor, and visual cortices of the felids, there were also notable differences. The African lion had significantly larger soma measures across all three cortical regions. Pyramidal neuron soma parameters were also largest in the prefrontal cortex of the African lion. Comparative studies in primates have revealed that pyramidal neurons in the prefrontal lobe tend to be more complex than those found in other cortical regions, indicative of the greater integrating abilities of the prefrontal lobe and its role in cognition and memory (Elston, 2000; Jacobs et al., 2001). In the current analyses, the soma parameters align well with this pattern of increased dendritic complexity in lion prefrontal cortex relative to other cortical regions. The overall larger soma parameters observed in the prefrontal cortex of the African lion (in comparison to the cheetah and leopard), might be related to the unique sociality of this species. Lions are the most gregarious of wild felids, with female lions maintaining social links throughout their lives (VanderWaal, Mosser, & Packer, 2009) and communally rearing young (Packer & Pusey, 1997). It is likely that the cognitive demands of social living in lions have brought about similar changes towards increased morphological complexity in the prefrontal cortex as described in primates (Hauser, 1999). Finally, gigantopyramidal soma parameters between the African lion and cheetah were similar, providing tentative evidence of morphological convergence. Observations based on regression analysis underscore the broad similarities within Carnivora, with the cheetah, African lion, and African leopard soma parameters scaling as predicted based on brain size. These observations highlight the order specific scaling attributes shared among the Carnivora.

5 | CONCLUSION

The present findings regarding somatodendritic morphology of cortical neurons in felids, to the extent that one can generalize from such case studies, contribute to a more comprehensive understanding of neuronal diversity in mammal brain evolution. In general, the neuronal morphology of the African lion, African leopard, and cheetah were descriptively consistent with what has been observed in the domestic cat and in other large felids. Quantitatively, pyramidal-like neurons in the cheetah were generally larger and more dendritically complex than in the African leopard despite similar brain masses. When comparing across multiple felids, the dendritic measures of typical pyramidal neurons generally increased with larger brain sizes for *Panthera* and closely-related species such that: Siberian tiger > African lion > African leopard > clouded leopard. Pyramidal neurons in the cheetah and the caracal, however, did not follow this pattern and had more complex dendrites than predicted. Finally, in accordance with Jacobs et al. (2018), the current Golgi and stereological analyses indicate that M1 gigantopyramidal neurons in the African lion, African leopard, and cheetah are exceptionally large for their relative brain masses. The large size of M1 gigantopyramidal neurons in the limited number of feliforms examined to date, particularly in the evolutionarily divergent non-*Panthera* cheetah, suggests that these neuronal types may be an example of convergent evolution within Felidae and may innervate specialized musculoskeletal systems designed for felid-specific hunting behavior.

Although examination of single animal within a species allows for a basic classification of neuronal morphology, a larger number of animals within each species is necessary for more comprehensive and quantitative comparisons. To this end, metadata obtained from online archives of digital reconstructions provide a platform for exploration of neuronal morphology, both within species (Hansen, Jespersen, Leigland, & Kroenke, 2013) and across species (Parekh, Armañanzas, & Ascoli, 2015; Parekh & Ascoli, 2015; Polavaram, Gillette, Parekh, & Ascoli, 2014). The Jacobs archive in neuromorpho.org, for example, had over one million downloads of traced neurons in 2015 (Ascoli, 2015) and, at the time of this publication, over two million. As noted by Manton et al. (2019), who recently examined 3,174 neurons across 24 different species from the Jacobs neuromorpho.org archive, it is particularly important for cross-species comparisons that neurons be acquired with the same methodology. Beyond morphology, other studies have examined dendritic computation patterns and topological factors across species using the same Jacobs archive (Kirch & Gollo, 2019). Thus, investigations such as the present one, although limited in scope, contribute to a larger database, which facilitates more sophisticated neuronal comparisons across species.

CONFLICT OF INTEREST

The authors have no conflicts of interest.

AUTHOR CONTRIBUTIONS

All authors had full access to all data in the study and take responsibility for the integrity of the data and the accuracy of the data analysis. P.R.M., C.C.S., P.R.H., B.J., M.S., and V.T.N. involved in study concept and design. V.T.N., R.U., A.W., and L.J.S. involved in collection and qualitative analysis of data. M.S., B.J., M.S.S., and M.A.S. involved in statistical analysis and interpretation. B.W., K.B., P.R.M., C.C.S., T.H., M.F.B., M.A.S., and C.D.S. involved in procurement, preparation, and fixation of tissue. P.R.M., P.R.H., C.C.S., B.J., and M.A.S. obtained

funding. B.J., M.S., P.R.M., P.R.H., M.A.S., C.C.S., and V.T.N. involved in drafting of the manuscript. P.R.M., R.U., A.W., B.J., M.S., C.C.S., M.A.S., V.T.N., and L.J.S. involved in photomicrography and preparation of figures/tables. V.T.N., P.R.M., P.R.H., C.C.S., M.A.S., and B.J. involved in critical revision of the manuscript for important intellectual content. B.J., P.R.H., P.R.M., C.C.S., and M.A.S. involved in study supervision.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are openly available http://neuromorpho.org/ under the corresponding author's name (Jacobs).

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